► BUILDING CODES
- DESIGN AND CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE PROVISIONS OF THE 2021 WISCONSIN COMMERCIAL BUILDING CODE (BASED ON IBC 2015) AS CONTAINED IN CHAPTERS SPS 361, SPS 362 and SPS 366 OF THE WISCONSIN ADMINISTRATIVE CODE.

RISK CATEGORY	II

► DESIGN LOADS AND DATA
- SOIL DATA (AGGREGATE BACKFILL)

ACTIVE SOIL PRESSURE	40 PSF PER FOOT OF DEPTH
AT-REST SOIL PRESSURE	65 PSF PER FOOT OF DEPTH
PASSIVE SOIL PRESSURE	360 PSF PER FOOT OF DEPTH
SURCHARGE LOAD	120 PSF

- ROOF LOADS

DEAD (7 PSF FRAMING, 2 PSF ROOFING, 12/15/18 PSF BALLAST* 4 PSF MEP)	25 TO 31 PSF (23 TO 29 PSF TO JOISTS)
* HEAVIER BALLAST AT 16' PERIMETER	
LIVE (SEE SNOW LOAD ALSO)	20 PSF

- SNOW LOADS

GROUND SNOW (p_g)	30 PSF
SNOW DENSITY (0.13 x p_g + 14 < 30)	17.9 PCF
ROOF EXPOSURE	PARTIALLY EXPOSED
EXPOSURE FACTOR (C_e)	1.0
THERMAL FACTOR - BUILDING (C_t)	1.0
THERMAL FACTOR - CANOPY, MANSARDS (C_t)	1.2
SNOW IMPORTANCE FACTOR (I_s)	1.0
FLAT ROOF SNOW LOAD - BUILDING	21 PSF (25 PSF USED)
FLAT ROOF SNOW LOAD - CANOPY	26 PSF
DRIFT LOAD	AS NOTED ON 1/S0.2 & DRAWINGS, PER ASCE 7-16

- WIND DATA

ULTIMATE DESIGN WIND SPEED - 3 SECOND GUST (V_{ULT})	107 MPH
NOMINAL DESIGN WIND SPEED - 3 SECOND GUST (V_{ASD})	83 MPH
BUILDING ENCLOSURE	ENCLOSED
EXPOSURE	C
WIND DIRECTIONALITY FACTOR (K_d)	0.85
TOPOGRAPHIC FACTOR (K_{zt})	1.0
GUST FACTOR (G - BUILDING IS RIGID)	0.85
INTERNAL PRESSURE COEFFICIENT (ENCLOSED - GC_{pi})	± 0.18
ANALYSIS PROCEDURE	DIRECTIONAL
EDGE ZONE WIDTH (2a)	28.3 FT
MEAN ROOF HEIGHT (h)	35.3 FT
ROOF PLANE SLOPE (Θ)	1.2 DEGREES
COMPONENTS AND CLADDING	SEE TABLE THIS SHEET
MINIMUM NET UPLIFT	
INTERIOR SPACES	10 PSF
EXTERIOR CANOPIES/SOFFITS	30 PSF

- SEISMIC DATA

SEISMIC IMPORTANCE FACTOR	1.00
MAPPED SPECTRAL RESPONSE ACCELERATION FOR SHORT PERIODS (S_S)	0.092
MAPPED SPECTRAL RESPONSE ACCELERATION FOR 1 SECOND PERIOD (S_1)	0.054
SITE CLASS PER GEOTECHNICAL REPORT	D
DESIGN SPECTRAL RESPONSE ACCELERATION FOR SHORT PERIODS (S_{DS})	0.098
DESIGN SPECTRAL RESPONSE ACCELERATION FOR 1 SECOND PERIOD (S_{D1})	0.086
SEISMIC DESIGN CATEGORY	B
BASIC SEISMIC FORCE RESISTING SYSTEM AND PARAMETERS	
BEARING WALL SYSTEM - ORDINARY PRECAST SHEAR WALLS	
R = 3.0 Ω_0 = 2.0 PRECAST C_d = 3.0	
SEISMIC RESPONSE COEFFICIENT (C_s)	0.033
ANALYSIS PROCEDURE	EQUIVALENT LATERAL FORCE (ASCE 12.8)

- RAIN

RAIN INTESNSITY	3.05 IN/HR
STATIC HEAD	6 IN
OVERFLOW DEVICE	24" LONG CHANNEL SCUPPER
HYDRAULIC HEAD	4.90 IN (MAX)
DESIGN RAIN LOAD (@ PROXIMITY DRAIN)	67.1 PSF (MAX)

► MATERIAL STRENGTHS AND STANDARDS
- SOILS

DESIGN SOIL BEARING CAPACITY FOR SPREAD/STRIP FOOTINGS	4,000 PSF
MODULUS OF SUB-GRADE REACTION	150 PCI
COEFFICIENT OF SLIDING FRICTION (CONCRETE –SOIL)	0.40

- REINFORCING STEEL

WELDED WIRE FABRIC, PROVIDED IN FLAT SHEETS ONLY (ASTM A185)	F_y = 65,000 PSI
DEFORMED BARS (ASTM A615, GRADE 60)	F_y = 60,000 PSI

- MASONRY

CONCRETE MASONRY ASSEMBLY (RUNNING BOND)	f'_m = 2,500 PSI
CMU (ASTM C90 - NORMALWEIGHT)	3,250 PSI
MORTAR (ASTM C270)	TYPE S
GROUT (ASTM C476)	f'_c = 3,000 PSI

- STRUCTURAL STEEL (SHAPES)

W, WT SECTIONS, CHANNELS (ASTM A992)	F_y = 50,000 PSI; F_u = 65,000 PSI
HP SECTIONS(ASTM A572, GRADE 50)	F_y = 50,000 PSI; F_u = 65,000 PSI
M, MT, S, ST SECTIONS, ANGLES (ASTM A36)	F_y = 36,000 PSI; F_u = 58,000 PSI
HSS SHAPES – SQUARE/RECTANGULAR (ASTM A500, GRADE C)	F_y = 50,000 PSI; F_u = 62,000 PSI
HSS SHAPES – ROUND (ASTM A500, GRADE C)	F_y = 50,000 PSI; F_u = 62,000 PSI
STEEL PIPE (ASTM A53, GRADE B)	F_y = 35,000 PSI; F_u = 60,000 PSI
PLATES (ASTM A36)	F_y = 36,000 PSI; F_u = 58,000 PSI

- STRUCTURAL STEEL (CONNECTIONS)

ANCHOR RODS (ASTM F1554, GRADE 36)	F_y = 36,000 PSI; F_u = 58,000 PSI
HIGH STRENGTH BOLTS (F3125)	GR. A325 UNLESS NOTED ON DRAWINGS
TENSION CONTROL BOLTS (F3125)	GR. F1852 UNLESS NOTED ON DRAWINGS
WELDING ELECTRODES	E70XX
SHEAR CONNECTORS (ASTM A29, GRADE 1010 THROUGH 1020; AWS TYPE B)	F_y = 51,000 PSI; F_u = 65,000 PSI
CONCRETE ANCHORS (ASTM A29, GRADE 1010 THROUGH 1020; AWS TYPE B)	F_y = 51,000 PSI; F_u = 65,000 PSI
DEFORMED BAR ANCHORS (ASTM A496; AWS TYPE C)	F_y = 70,000 PSI; F_u = 80,000 PSI
THREADED RODS (ASTM A36)	F_y = 36,000 PSI; F_u = 58,000 PSI
GROUT (ASTM C1107)	f'_c = 5,000 PSI

- COLD-FORMED METAL FRAMING

COLD-FORMED MATERIAL - 18 GAUGE AND THINNER (ASTM A653, GRADE 33)	F_y = 33,000 PSI
COLD-FORMED MATERIAL - 16 GAUGE AND THICKER (ASTM A653, GRADE 50)	F_y = 50,000 PSI
ANCHOR RODS (ASTM F1554, GRADE 36)	F_y = 36,000 PSI; F_u = 58,000 PSI
CONNECTOR PLATES (ASTM A36)	F_y = 36,000 PSI; F_u = 58,000 PSI
CONNECTOR BOLTS (ASTM A307, GRADE A)	F_u = 60,000 PSI
WELDING ELECTRODES	E60XX
GALVANIZING THICKNESS	G60

► GENERAL NOTES
- CONSTRUCTION

UNLESS SPECIFICALLY NOTED OTHERWISE, BUILDING STRUCTURE HAS BEEN DESIGNED FOR THE FINAL COMPLETED CONDITION ONLY, AND HAS NOT BEEN ANALYZED, INVESTIGATED OR DESIGNED FOR OVERALL STRUCTURE, OR INDIVIDUAL MEMBER, STABILITY DURING CONSTRUCTION. CONTRACTOR SHALL PROVIDE AND MAINTAIN TEMPORARY BRACING AND SUPPORTS FOR ALL STRUCTURAL ELEMENTS, BOTH INDIVIDUALLY AND COLLECTIVELY, AS REQUIRED AT EVERY STAGE OF CONSTRUCTION UNTIL THE FINAL COMPLETION OF THE STRUCTURE. NO PORTION OF THE BUILDING STRUCTURE, WHILE UNDER CONSTRUCTION IS INTENDED TO BE STABLE IN THE ABSENCE OF THE CONTRACTORS TEMPORARY BRACES AND SUPPORTS, WHICH SHALL ADDITIONALLY PROVIDE SUPPORT FOR ALL CONSTRUCTION LOADING. MATERIALS AND EQUIPMENT SHALL BE STORED, TRANSPORTED AND INSTALLED IN A MANNER THAT WILL NOT EXCEED THE DESIGN FLOOR LOADING.

CONTRACTOR IS SOLELY RESPONSIBLE FOR THE MEANS, METHODS, TECHNIQUES, SEQUENCES AND PROCEDURES OF CONSTRUCTION INCLUDING, BUT NOT LIMITED TO, TEMPORARY BRACING, SUPPORTS, SHORING, FORMING TO SUPPORT IMPOSED CONSTRUCTION LOADS, AND OTHER SIMILAR ITEMS.

STRUCTURAL DOCUMENTS MAY REFER TO OSHA REQUIREMENTS. SUCH REFERENCES ARE INCIDENTAL, AND ARE NOT INTENDED TO IDENTIFY ALL APPLICABLE OSHA REQUIREMENTS.

© 2026 Partners in Design Architects, Inc.

► GENERAL NOTES (CONTINUED)
- COMPLETENESS

INFORMATION CONTAINED IN THE GENERAL NOTES IS ONLY A PARTIAL SUMMARY OF PROJECT REQUIREMENTS. SEE SPECIFICATIONS, PLANS AND DETAILS FOR ADDITIONAL REQUIREMENTS.

USE ONLY DIMENSIONS INDICATED ON THE DRAWINGS. DO NOT MANUALLY SCALE THE DRAWINGS OR USE ANY DIMENSIONS MEASURED FROM ELECTRONIC DRAWING FILES.

UNLESS NOTED OTHERWISE, CENTERLINE OF FLOOR FRAMING ELEMENTS COINCIDES WITH COLUMN CENTERLINES, AND FRAMING ELEMENTS ARE EQUALLY SPACED BETWEEN ADJACENT COLUMN CENTERLINES.

MAJOR OPENING LOCATIONS AND SIZES ARE INDICATED ON THE STRUCTURAL DRAWINGS - SMALLER OPENINGS AND SLEEVES REQUIRED TO ACCOMMODATE VARIOUS BUILDING SERVICES MAY NOT BE NOTED. CONTRACTOR TO VERIFY THE SIZE AND LOCATION OF ALL ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING OPENINGS, INCLUDING CLEARANCE REQUIREMENTS CONTAINED IN THE RESPECTIVE DISCIPLINE DOCUMENTS FOR INSTALLATION AND IN-PLACE OPERATION OF THE RESPECTIVE EQUIPMENT OR ITEMS.

CONSULT ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING DRAWINGS AND MANUFACTURERS SPEC SHEETS FOR LOCATIONS AND DIMENSIONS OF PADS, CURBS, EQUIPMENT SUPPORTS, DEPRESSIONS, INSERTS, DRIPS, REGLETS, REVEALS, FINISHES AND OTHER MISCELLANEOUS PROJECT REQUIREMENTS THAT NECESSITATE INCIDENTAL ACCOMMODATION BY THE BUILDING STRUCTURE BUT ARE NOT SHOWN ON THE STRUCTURAL DRAWINGS.

DEFERRED SUBMITTALS: IBC SECTION 107.3.4.1 REQUIRED DEFERRED SUBMITTALS TO BE APPROVED BY THE BUILDING OFFICIAL. DEFERRED SUBMITTALS SHALL BE REVIEWED BY DESIGN PROFESSIONAL IN RESPONSIBLE CHARGE AFFIXING A NOTATION THAT THE DEFERRED ITEMS ARE IN GENERAL CONFORMANCE WITH THE DESIGN OF THE BUILDING, SUBMITTED FOR REVIEW BY THE BUILDING OFFICIAL AND SHALL NOT BE INSTALLED UNTIL APPROVED BY THE BUILDING OFFICIAL. DEFERRED SUBMITTALS ON THIS PROJECT INCLUDE:
- COLD FORMED METAL FRAMING (DFMF)
- PRECAST WALL PANELS
- STAIRS/ LADDERS/ RAILINGS

- GENERAL

THE STRUCTURE HAS BEEN DESIGNED AS UNRESTRAINED FOR THE PURPOSE OF FIRE RATING AND FIREPROOFING ASSEMBLY EVALUATIONS.

STRUCTURAL COMPONENTS HAVE NOT BEEN DESIGNED FOR VIBRATORY EQUIPMENT UNLESS NOTED OTHERWISE. PLACE VIBRATORY EQUIPMENT AND EQUIPMENT SENSITIVE TO VIBRATIONS ON VIBRATION ISOLATORS SPECIFICALLY DESIGNED FOR THE EQUIPMENT.

ALL SYSTEMS, INCLUDING EXTERIOR FACADES AND FRAMING, WHICH ARE DESIGNED AND DETAILED BY COMPONENT SUPPLIERS, ARE ASSUMED TO IMPOSE VERTICAL AND/OR HORIZONTAL LOADS ON THE BASE BUILDING STRUCTURE WITHOUT CAUSING TORSION IN THE SUPPORTING STRUCTURAL MEMBERS. COMPONENT SUPPLIERS ARE RESPONSIBLE FOR DESIGNING, FURNISHING AND INSTALLING SUPPLEMENTARY BRACING MEMBERS AS REQUIRED TO PREVENT THEIR SYSTEMS FROM CAUSING TORSION IN THE SUPPORTING STRUCTURAL MEMBERS. WHERE PROVIDED, SUPPLEMENTARY BRACING SHALL NOT INTERFERE WITH ANY BUILDING SYSTEM NOTED OR DESCRIBED IN THE CONTRACT DOCUMENTS.

UNDER NO CIRCUMSTANCES MAY ANY STRUCTURAL ELEMENT BE PENETRATED, CUT, NOTCHED, BLOCKED-OUT, SLEEVED, CORE DRILLED, OR OTHERWISE FIELD MODIFIED OR REDUCED IN STRENGTH AFTER DELIVERY TO THE CONSTRUCTION SITE OR FINAL INCORPORATION IN THE BUILDING STRUCTURE UNLESS SUCH MODIFICATION IS SPECIFICALLY DETAILED ON THE STRUCTURAL DRAWINGS, OR IS APPROVED IN ADVANCE IN WRITING BY THE ENGINEER OF RECORD.

ANY FUTURE ADDITION IS ASSUMED TO BE COMPLETELY SELF SUPPORTING FOR BOTH GRAVITY AND LATERAL LOADS.

► SYSTEM NOTES
- FOUNDATIONS AND EARTHWORK

REMOVE EXISTING SURFICIAL TOP SOIL AND VEGETATION FROM WITHIN THE BUILDING AREA AND A MINIMUM OF TEN FEET BEYOND. EXCAVATE MATERIAL TO PROPOSED SLAB-ON-GROUND SUBGRADE. PROOFROLL WITH A HEAVY RUBBER TIRED VEHICLE. SOILS WHICH HEAVE, PUMP, OR DO NOT READILY COMPACT SHALL BE EXCAVATED AND REPLACED WITH ENGINEERED FILL.

SUBGRADE PREPARATION FOR FOOTINGS SHALL CONSIST OF EXCAVATION TO REQUIRED ALLOWABLE BEARING CAPACITY SOILS AT OR NEAR DESIGN FOOTING ELEVATIONS. WHERE UNSUITABLE SOIL IS ENCOUNTERED AT NOMINAL BEARING DEPTH, SEE OVER EXCAVATION DETAIL.

ALL COMPACTION REQUIREMENTS REFER TO % OF MAXIMUM DRY DENSITY PER ASTM D-1557 MODIFIED PROCTOR. GRANULAR STRUCTURAL FILL BENEATH FOOTINGS SHALL BE PLACED IN LAYERS NO MORE THAN 8" THICK, AND EACH LAYER SHALL BE COMPACTED TO 95%. COHESIVE FILL APPROVED BY THE GEOTECHNICAL CONSULTANT SHALL BE PLACED IN LAYERS NO THICKER THAN 8", AND EACH LAYER SHALL BE COMPACTED TO 95%. MOISTURE CONDITION FILL MATERIALS AS REQUIRED TO OBTAIN PROPER COMPACTION. COHESIVE SOILS OR GRANULAR SOILS WITH A SIGNIFICANT PERCENT OF COHESIVE FINES SHALL BE CONDITIONED TO WITHIN 3% OF OPTIMUM MOISTURE CONTENT AT COMPACTION.

FOR GENERAL INFORMATION AND SPECIFIC RECOMMENDATIONS AND REQUIREMENTS PERTAINING TO THE PROJECT SITE, REFER TO THE PROJECT GEOTECHNICAL REPORT PREPARED BY ECS MIDWEST, LLC PROJECT NUMBER 42:1291, DATED AUGUST 25, 2016.

ALL ACTIVITIES CONCERNING PREPARATION AND VERIFICATION OF BEARING SOILS FOR SLAB-ON-GROUND AND FOOTINGS SHALL BE SUPERVISED AND APPROVED BY A QUALIFIED GEOTECHNICAL ENGINEER.

COLUMNS, PIERS, AND SPREAD FOOTINGS ARE CENTERED ON GRIDLINES UNLESS NOTED OTHERWISE. CONTINUOUS FOOTINGS ARE CENTERED ON WALLS ABOVE UNLESS NOTED OTHERWISE.

BACKFILL UNIFORMLY ON EACH SIDE OF FOUNDATION WALLS, GRADE BEAMS AND OTHER SIMILAR ELEMENTS. DO NOT BACKFILL AGAINST ANY STRUCTURAL ELEMENT UNTIL THAT ELEMENT HAS ATTAINED FULL DESIGN STRENGTH. DO NOT BACKFILL AGAINST BASEMENT WALLS UNTIL TOP AND BOTTOM OF WALL IS BRACED BY FLOOR FRAMING AND SLAB-ON-GROUND UNLESS NOTED OTHERWISE ON THE DRAWINGS.

TOP OF FOOTING ELEVATION NOTED ON DRAWINGS REPRESENT CONSIDERED ENGINEERING JUDGMENT BY THE ENGINEER OF RECORD ABOUT PROTECTION FROM FROST AND MINIMUM DEPTH TO SOILS CAPABLE OF PROVIDING DESIGN SOIL BEARING CAPACITY. UNCERTAINTIES INHERENT IN DETERMINING THE ELEVATION OF SOILS ADEQUATE TO PROVIDE DESIGN BEARING CAPACITY MAY REQUIRE FOUNDATIONS TO BE LOWERED – IN NO CASE SHALL TOP OF FOOTING BE HIGHER THAN NOTED. A GEOTECHNICAL ENGINEER SHALL VERIFY THAT SOIL AT THE FOOTING BASE IS ADEQUATE TO PROVIDE THE REQUIRED DESIGN SOIL BEARING CAPACITY.

- CAST-IN-PLACE CONCRETE

DESIGN AND CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE PROVISIONS OF ACI 318–18 USING STRENGTH DESIGN METHODOLOGY, EXCEPT WHERE MORE RESTRICTIVE REQUIREMENTS ARE NOTED.

REINFORCING CLEAR COVER SHALL BE AS NOTED BELOW UNLESS SPECIFICALLY NOTED OTHERWISE ON STRUCTURAL DRAWINGS.

CONCRETE CAST AGAINST AND PERMANENTLY EXPOSED TO EARTH	3"
CONCRETE EXPOSED TO EARTH OR WEATHER	
#3 - #5 BARS	1 1/2"
#6 - #18 BARS	2"
CONCRETE NOT EXPOSED TO EARTH OR WEATHER	
WALLS - #3 THRU #11 BARS	3/4"
WALLS - #14 THRU #18 BARS	1 1/2"
STRUCTURAL SLABS - TOP, BOTTOM	1"
JOIST TIES AND MAIN REINFORCING - TOP, BOTTOM , SIDES	1 1/2"
BEAM TIES - TOP, BOTTOM, SIDES	1 1/2"
BEAM MAIN REINFORCING - TOP, BOTTOM, SIDES	2"
COLUMN TIES	1 1/2"
COLUMN MAIN REINFORCING	2"

PROVIDE (2) #5 BARS AROUND ALL OPENINGS AND (2) #5 DIAGONAL BARS AT ALL OPENING AND RE-ENTRANT CORNERS. BARS SHALL EXTEND A MINIMUM OF 24" PAST OPENING.

ALL BAR SPLICES SHALL BE CONTACT LAP SPLICED USING CLASS B TENSION LAP LENGTHS, WITH ADJACENT LAPS STAGGERED A MINIMUM OF 3'-0" UNLESS DETAILED OTHERWISE. SEE REINFORCEMENT TABLES FOR REQUIRED LAP AND DEVELOPMENT LENGTHS.

FIELD WELDING OF ASTM A615 REINFORCING STEEL IS NOT PERMITTED. FIELD BENDING OF REINFORCING STEEL IS NOT PERMITTED EXCEPT WHERE SPECIFICALLY DETAILED ON STRUCTURAL DRAWINGS.

CORING OF COLUMNS, WALLS, BEAMS, JOISTS AND SLABS IS NOT PERMITTED. PROVIDE STEEL SLEEVES FOR ALL PENETRATIONS AT ALL LOCATIONS APPROVED BY THE ENGINEER OF RECORD PRIOR TO PLACING CONCRETE.

- PRECAST CONCRETE

DESIGN AND CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE PROVISIONS OF ACI 318–14 AND THE PCI DESIGN HANDBOOK, SEVENTH EDITION, USING STRENGTH DESIGN METHODOLOGY, EXCEPT WHERE MORE RESTRICTIVE REQUIREMENTS ARE NOTED.

PRECAST MEMBERS SHALL BE DESIGNED TO SUPPORT THEIR OWN SELF WEIGHT, THE SUPERIMPOSED LOADS NOTED ON THE STRUCTURAL PLANS AND DETAILS, AND TO ACCOMMODATE THE DETAILS AND ADDITIONAL LOADS THAT MAY BE SHOWN ON ARCHITECTURAL, MECHANICAL, PLUMBING OR ELECTRICAL PLANS.

► SYSTEM NOTES (CONTINUED)
- CONCRETE MASONRY

DESIGN AND CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE PROVISIONS OF TMS 402-13 AND TMS 602-13 EXCEPT WHERE MORE RESTRICTIVE REQUIREMENTS ARE NOTED.

UNLESS NOTED OTHERWISE PROVIDE CONTINUOUS LADDER TYPE REINFORCEMENT WITH W1.7 (9 GAUGE) SIDE AND CROSS RODS AT 16" OC VERTICALLY IN ALL WALLS AND PIERS, AND AT 8" OC VERTICALLY AT PARAPETS. WHERE VERTICAL BARS ARE REQUIRED, CONSTRUCT CMU WALL TO PROVIDE A CONTINUOUS UNOBSTRUCTED CELL FROM BOTTOM TO TOP OF BAR. CELL CONTAINING A SINGLE BAR SHALL NOT BE LESS THAN 3" X 4" IN PLAN AREA.

PORTIONS OF CMU CONSTRUCTION REQUIRING STRUCTURAL FILL SHALL USE GROUT ONLY. USE OF CONCRETE FILL IN CMU CONSTRUCTION IS NOT PERMITTED. WHERE CLEARANCES AND CONGESTION PERMIT, USE COARSE GROUT WITH PEA GRAVEL AGGREGATE; OTHERWISE USE FINE GROUT.

REFER TO ARCHITECTURAL DRAWINGS FOR LOCATION OF ALL VERTICAL CONTROL JOINTS IN EXTERIOR WYTHES OF EXTERIOR WALLS, AND IN ALL INTERIOR PARTITION WALLS.

PROVIDE STEEL PIPE SLEEVES AT ALL LOCATIONS WHERE PIPING PASSES THROUGH CMU WALL.

WHERE BOND BEAMS INTERSECT AT WALL CORNERS AT DIFFERENT ELEVATIONS, RUN EACH BOND BEAM AROUND THE CORNER FOR A MINIMUM OF TWO FULL BLOCK LENGTHS BEFORE TERMINATING. WHERE BOND BEAMS ADJOIN ON THE SAME WALL AT DIFFERENT ELEVATIONS, RUN BOND BEAMS PAST ONE ANOTHER A MINIMUM OF FOUR FULL BLOCK LENGTHS BEFORE TERMINATING.

- STRUCTURAL STEEL

DESIGN, DETAILING, AND CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE SPECIFICATION FOR STRUCTURAL STEEL BUILDINGS (AISC 360-16), THE CODE OF STANDARD PRACTICE FOR STEEL BUILDINGS AND BRIDGES (303-16), AND THE STEEL CONSTRUCTION MANUAL FIFTEENTH EDITION.

TYPICAL DETAILS INDICATE GENERAL CRITERIA FOR DESIGN AND DETAILING OF CONNECTIONS. THEY ARE NOT INTENDED TO CONVEY COMPLETE INFORMATION CONCERNING SIZE AND QUANTITY OF CONNECTORS, PLATES, ANGLES, WELDS AND SIMILAR ITEMS THAT ARE DEVELOPED THROUGH THE DESIGN OF AN INDIVIDUAL CONNECTION FOR A SPECIFIC SET OF LOADS AND COMBINATIONS. DETAILS THAT CONVEY SPECIFIC COMPONENT INFORMATION ESTABLISH MINIMUM REQUIREMENTS AND ARE NOT INTENDED TO CONVEY A COMPLETE DESIGN UNLESS NOTED.

UNLESS OTHERWISE NOTED, ALL STEEL TO STEEL FRAMING HAS BEEN SELECTED ASSUMING ATTACHMENTS FOR SHEAR ONLY, USING DOUBLE ANGLE OR DOUBLE BENT PLATE CONNECTIONS SHOP WELDED TO FRAMING MEMBER AND FIELD BOLTED TO SUPPORTING MEMBER WITH HIGH STRENGTH BOLTS IN BEARING. CONNECTIONS SHALL BE SYMMETRICAL ABOUT THE BEAM WEB. FABRICATORS PROPOSING TO USE ALTERNATIVE METHODS OF ATTACHMENT NOT SPECIFICALLY DETAILED ON THE STRUCTURAL DRAWINGS SHALL SUBMIT ALTERNATIVE FOR CONSIDERATION DURING BIDDING, AND SHALL BEAR ALL COSTS ASSOCIATED WITH REVIEW, ENGINEERING REDESIGN, AND APPROVAL OF ALTERNATIVE CONNECTIONS.

SINGLE PLATE SHEAR TAB CONNECTIONS MAY BE USED IN LIEU OF DOUBLE ANGLE OR DOUBLE BENT PLATE CONNECTIONS WHERE SPECIFICALLY NOTED ON DRAWINGS. TO HSS OR PIPE COLUMNS, OR WHERE CONNECTION OF FRAMING MEMBER TO ONE SIDE OF A SUPPORT MEMBER IS MATCHED BY A SIMILAR CONNECTION ON THE OPPOSITE SIDE OF THE SAME SUPPORT MEMBER, AND WHERE BEAM SPANS DO NOT DIFFER BY MORE THAN 50% OF THE LARGER SPAN. SINGLE PLATE SHEAR TABS MAY NOT BE USED FOR CONNECTION OF FRAMING MEMBERS TO WIDE FLANGE COLUMNS OR TO SPANDREL (EDGE) SUPPORT MEMBERS UNLESS SPECIFICALLY DETAILED ON DRAWINGS.

FOR ALL SLOTTED HOLES, ANCHORS SHALL BE CENTERED IN SLOT UNLESS NOTED OTHERWISE.

CONNECTIONS FOR ALL STRUCTURAL STEEL BEAMS AND GIRDERS SHALL BE DESIGNED FOR THE REACTIONS SHOWN. REACTIONS SHOWN ON THE DRAWINGS WITHOUT INDIVIDUAL LOAD DESIGNATIONS HAVE BEEN COMBINED USING STRENGTH COMBINATIONS. IF NO REACTION IS SHOWN:
- NON-COMPOSITE BEAM AND GIRDER SHEAR CONNECTIONS SHALL BE DESIGNED FOR 50% OF THE TOTAL UNIFORM LOAD CAPACITY FOR THE GIVEN MEMBER SIZE, SPAN, AND GRADE OF STEEL.
- UNLESS DESIGN MOMENTS ARE SPECIFICALLY NOTED ON THE DRAWINGS, MOMENT CONNECTIONS SHALL BE DESIGNED TO FULLY DEVELOP THE MOMENT CAPACITY OF THE BEAM OR GIRDER

CONNECTIONS NOT SHOWN OR NOT COMPLETELY DETAILED ON THE DRAWINGS SHALL BE COMPLETED BY ONE OF THE FOLLOWING METHODS:
- STANDARD AISC FRAMED CONNECTIONS MEETING OTHER REQUIREMENTS OF THE DRAWINGS AND SPECIFICATIONS MAY BE SELECTED OR COMPLETED BY AN EXPERIENCED STEEL DETAILER RETAINED BY THE FABRICATOR IN ACCORDANCE WITH AISC 303-16, PARAGRAPH 3.1.1.3 AND USING STRENGTH DESIGN CONCEPTS.
- ALL CONNECTIONS IN THE FOLLOWING LIST AND ALL CONNECTIONS NOT MEETING THE LIMITATIONS OF STANDARD AISC FRAMED CONNECTIONS SHALL BE DESIGNED BY A PROFESSIONAL ENGINEER REGISTERED IN THE STATE IN WHICH THE PROJECT IS CONSTRUCTED AND RETAINED BY THE FABRICATOR IN ACCORDANCE WITH AISC AISC 303-16, PARAGRAPH 3.1.1.3 AND USING STRENGTH DESIGN CONCEPTS:
  1. TENSION CONNECTIONS NOT COMPLETELY DETAILED ON THE DRAWINGS, INCLUDING CONSIDERATION OF PRYING ACTION.
  2. MOMENT CONNECTIONS NOT COMPLETELY DETAILED ON THE DRAWINGS.
  3. BRACED FRAME CONNECTIONS NOT COMPLETELY DETAILED ON THE DRAWINGS

IN NO CASE SHALL A CONNECTING ELEMENT CONTAIN LESS THAN TWO HIGH STRENGTH BOLTS ON EACH CONNECTING SURFACE, OR UTILIZE LESS THAN 3/16" FILLET WELDS, OR BE LESS THAN T/2 IN DEPTH. IN NO CASE SHALL A CONNECTION BE DESIGNED FOR A SHEAR REACTION LESS THAN 18 KIPS (STRENGTH DESIGN)

DESIGN OF STAIRS, HANDRAILS, GUARDRAILS AND LADDERS SHALL BE BY THE STEEL SUPPLIER.

REFER TO ARCHITECTURAL DRAWINGS FOR MISCELLANEOUS STRUCTURAL STEEL NOT NOTED ON STRUCTURAL DRAWINGS.

PROVIDE CLEVISES, PINS, RODS AND TURNBUCKLES WITH CAPACITIES NOT LESS THAN THOSE LISTED IN THE AISC STEEL CONSTRUCTION MANUAL.

PROVIDE HOLES IN BEAMS TO ACCOMMODATE WOOD CONNECTIONS TO STEEL.

WELDER REQUIREMENTS: TECHNIQUE FOR WELDING AND QUALIFICATION OF WELDING PERSONNEL SHALL BE IN ACCORDANCE WITH THE SPECIFICATIONS LISTED IN IBC SECTION 2205, 2206, 2207, 2208, 2210, AND 2211 AS APPLICABLE. ALL WELDING IS TO BE COMPLETED BY A LICENSED STRUCTURAL WELDER ISSUED BY GOVERNING AUTHORITY IN WHICH STATE THE PROJECT RESIDES.

- JOISTS

DESIGN, FABRICATION AND ERECTION OF JOISTS AND LONGSPAN JOISTS, AND JOIST GIRDERS SHALL BE IN ACCORDANCE WITH THE 2015 STEEL JOIST INSTITUTE (SJI) STANDARDS

ALL STANDARD K AND LH SERIES JOISTS SHALL BE DESIGNED FOR A SHEAR CAPACITY EQUAL TO THE REACTION, AND VARYING LINEARLY TO 25% OF THE REACTION AT THE MIDSPAN OF THE JOIST. IN ORDER TO ACCOUNT FOR POTENTIAL STRESS REVERSALS THE SHEAR CAPACITY OF THE JOIST SHALL BE MAINTAINED AT THE 25% VALUE FOR A DISTANCE BEYOND THE MIDSPAN EQUAL TO MINIMUM OF ONE PANEL WIDTH, ROUNDED UP TO THE NEXT PANEL POINT.

WHERE JOISTS ARE DESIGNATED BY DEPTH, SERIES AND TOTAL LOAD / LIVE LOAD, FINAL DESIGN SHALL BE PER NOTED LOAD PLUS SELF WEIGHT OF JOIST AND IS THE RESPONSIBILITY OF THE JOIST SUPPLIER.

WHERE STANDARD JOIST DESIGNATION FOR DEPTH AND SERIES PRECEDES THE DESIGNATION "SP", SPECIAL JOIST SHALL HAVE THAT DEPTH AND THE STANDARD SEAT DEPTH FOR THE SPECIFIED SERIES. JOIST SUPPLIER SHALL PROVIDE A DESIGN PER THE PROVIDED LOADING CONDITIONS PLUS THE SELF WEIGHT OF THE JOIST.

WHERE STANDARD JOIST DESIGNATION FOR DEPTH AND SERIES PRECEDES THE DESIGNATION "SP", FINAL DESIGN SHALL BE PER LOADING DIAGRAM PROVIDED PLUS SELF WEIGHT OF JOIST, SHALL USE THE DEPTH AND STANDARD SEAT DEPTH AND SERIES INDICATED AND IS THE RESPONSIBILITY OF THE JOIST SUPPLIER.

WHERE BRIDGING INTERFERES WITH MECHANICAL OR OTHER TRADE INSTALLATION, CONTRACTOR MAY REMOVE BRIDGING AFTER METAL DECK IS COMPLETE IN PLACE, UPON RECEIPT OF WRITTEN APPROVAL FROM THE JOIST ENGINEER. BRIDGING REMOVED SHALL BE REPLACED AS DIRECTED BY THE JOIST ENGINEER, INCLUDING ADDITIONAL SUPPLEMENTAL BRACING AS MAY BE NECESSARY IN THE SOLE JUDGEMENT OF THE JOIST ENGINEER.

NO FIELD DRILLED HOLES OR CUTS ARE PERMITTED IN ANY JOIST CHORD OR WEB MEMBER.

MAXIMUM HANGER LOAD TO BE LOCATED ALONG BAR JOIST TOP CHORD BETWEEN PANEL POINTS IS 350 POUNDS AND 100 POUNDS TO JOIST BOTTOM CHORD BETWEEN PANEL POINTS. JOIST SUPPLIER TO DESIGN CHORDS FOR THESE LOADS AS A BEND-CHECK CONDITION.

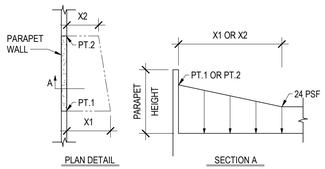
ALL CONCENTRATED LOADS EXCEEDING THE VALUES IN THE PRECEDING PARAGRAPH SHALL BE APPLIED AT A JOIST PANEL POINT UNLESS LOADS ARE INDICATED ON LOAD DIAGRAMS AND CHORDS HAVE BEEN SPECIFICALLY DESIGNED FOR CONCENTRATED LOADS, OR UNLESS SUPPLEMENTAL CHORD BRACING IS PROVIDED. SUPPLEMENTAL CHORD BRACING SHALL BE PROVIDED AS DETAILED ON THE DRAWINGS BY THE CONTRACTOR RESPONSIBLE FOR THE CONCENTRATED LOADS NOT APPLIED AT PANEL POINTS.

JOISTS AND SEAT CONNECTIONS SHALL BE DESIGNED TO RESIST AXIAL LOADS INDICATED, OR RESIST A HORIZONTAL FORCE ACTING PARALLEL TO THE JOIST NOT LESS THAN 5% OF THE (DEAD + LIVE) LO

CONCRETE MIX DESIGN

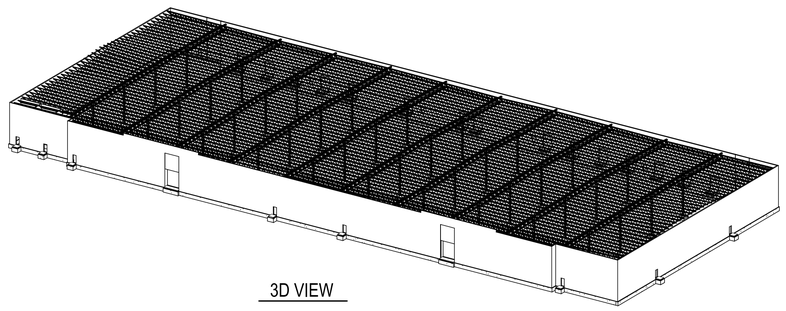
CLASS	TYPE OF CONSTRUCTION	SPECIFIED COMP STRENGTH AT 28 DAYS (PSI)	MAXIMUM AGGREGATE SIZE (IN)	MINIMUM LBS OF CEMENT PER CY (NOTE 7)	AIR ENTRAINMENT % +/- 1 1/2%	NOTES
1	ALL FOOTINGS	3,000	1.5	470	NONE	4
2	WALLS/PIERS	4,000	0.75	494	NONE	4
3	EXPOSED WALLS	4,000	0.75	540	6.0	5
4	INTERIOR SLAB-ON-GROUND	4,000	1.0 (NOTE 6)	540	NONE	3,4,8
5	EXTERIOR SLAB-ON-GROUND	4,500	0.75	564	6.0	1,2,5
6	BOND BEAMS	3,000	0.375	470	NONE	4
7	MISCELLANEOUS NON-SCHEDULED CONCRETE WORK	3,000	0.75	470	6.0	1,5

NOTES:
1. AIR ENTRAINED CONCRETE, USE FOR EXTERIOR WALLS, EXTERIOR SLABS, WALKS, PLATFORMS, RAMPS, STEPS, PORTIONS OF PARKING RAMP AND OTHER CONCRETE EXPOSED TO FREEZING AND THAWING.
2. MAXIMUM WATER—CEMENTITIOUS RATIO BY WEIGHT SHALL BE 0.45.
3. MAXIMUM WATER—CEMENTITIOUS RATIO BY WEIGHT SHALL BE 0.50.
4. A MAXIMUM OF 50 PERCENT TOTAL REPLACEMENT OF PORTLAND CEMENT WITH GGBFS (GROUND GRANULATED BLAST—FURNACE SLAG) AND FLY ASH AT A 1:1 RATIO; UP TO 350 POUNDS, WITH A MAXIMUM 25 PERCENT FLY ASH. IF FLY ASH IS USED ALONE, LIMIT MAXIMUM REPLACEMENT TO 25 PERCENT.
5. A MAXIMUM OF 30 PERCENT TOTAL REPLACEMENT OF PORTLAND CEMENT WITH GGBFS (GROUND GRANULATED BLAST—FURNACE SLAG) AND FLY ASH AT A 1:1 RATIO WHERE FREEZE—THAW DURABILITY AND EXPOSURE TO DEICERS IS LIKELY; UP TO 350 POUNDS, WITH A MAXIMUM 25 PERCENT FLY ASH. IF FLY ASH IS USED ALONE, LIMIT MAXIMUM REPLACEMENT TO 25 PERCENT.
6. INCREASE AGGREGATE SIZE TO 1 1/2" IF SLAB-ON-GROUND TO BE PLACED BY LASER SCREED .
7. MINIMUM AMOUNT OF CEMENTITIOUS MATERIAL IDENTIFIED IN THE MIX PROPORTIONS SHALL APPLY FOR MIXES FOR WHICH FIELD EXPERIENCE OR TRIAL MIXTURE INFORMATION REQUIRED IS NOT PROVIDED.
8. MIX SHALL BE DESIGNED SUCH THAT SHRINKAGE SHALL BE LESS THAN OR EQUAL TO 0.040 AFTER 28 DAYS. CONCRETE SUPPLIER SHALL PERFORM SHRINKAGE BEAM TESTING TO VERIFY CONCRETE PERFORMANCE IN ACCORDANCE WITH ASTM C157 AND SUBMIT RESULTS FOR REVIEW.



DRIFT MARK	PARAPET HT. PT.1	PARAPET HT. PT.2	DRIFTING SNOW LOAD PT.1 PSF	PT.1 X1	PT.2 PSF	PT.2 X2
A	2'-0"	1'-5"	35.8	4.8'	25	0'
B	2'-0"	2'-0"	35.8	4.8'	35.8	4.8'
C	4'-0"	3'-0"	71.6	20.7'	53.7	12.8'

1 S0.2 DRIFTING SNOW LOADS
SCALE: 12" = 1'-0"



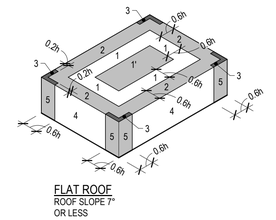
3D VIEW

COMPONENTS AND CLADDING WIND PRESSURES (PSF)

ZONE	WIND AREA (SF)	ROOF SLOPE 0° TO 7° (+)	(-)
1	10	8.9	34.7
1	50	7.6	29.4
1	100	7.0	27.1
1'	10	8.9	19.9
1'	50	7.6	19.9
1'	100	7.0	19.9
2	10	8.9	45.7
2	50	7.6	38.9
2	100	7.0	36.0
3	10	8.9	62.3
3	50	7.6	48.7
3	100	7.0	42.8
(-) WIND PRESSURE ON ROOF OVERHANGS			
LOCATION	WIND AREA (SF)	ROOF SLOPE 0° TO 7° ZONES 1&1'	ZONE 2	ZONE 3
OVERHANG	10	31.4	42.4	59.0
OVERHANG	20	30.8	38.5	52.2
OVERHANG	50	30.1	33.3	43.1
OVERHANG	100	29.5	29.4	36.2

ZONE	WIND AREA (SF)	(+)	(-)
4	10	21.8	23.6
4	20	20.8	22.6
4	50	19.5	21.3
4	100	18.5	20.4
5	10	21.8	29.1
5	20	20.8	27.2
5	50	19.5	24.6
5	100	18.5	22.6
ADJUSTMENT FACTOR (A)			
MEAN ROOF HEIGHT (FT)		EXPOSURE B	C	D
15		0.82	1.21	1.47
20		0.89	1.29	1.55
30		1.00	1.40	1.66
35		1.05	1.45	1.70
40		1.09	1.49	1.74
45		1.12	1.53	1.78
50		1.16	1.56	1.81
55		1.19	1.59	1.84
60		1.22	1.62	1.87

NOTES:
1. TABULATED LOADS ARE BASED ON ASCE 7-16 SIMPLIFIED PROVISIONS FOR ENCLOSED REGULAR-SHAPED BUILDINGS WITH THE FOLLOWING PARAMETERS: WIND SPEED = 110 MPH, MEAN ROOF HEIGHT = 30'-0", EXPOSURE B, K_{zt} = 1.0.
2. FOR DIFFERENT MEAN ROOF HEIGHTS OR EXPOSURES, TABULATED VALUES SHALL BE MULTIPLIED BY THE ADJUST FACTOR (A) CONTAINED WITHING THE ABOVE TABLE.
3. FOR WIND PRESSURES BELOW 16 PSF AFTER ALL ADJUSTMENT FACTORS HAVE BEEN TAKEN INTO ACCOUNT, A MINIMUM WIND PRESSURE OF 16 PSF SHALL BE USED FOR DESIGN. THOSE PRESSURES INDICATED BY ▭ IN THE ABOVE TABLE ARE THOSE THAT FALL BELOW THE MINIMUM VALUE BASED ON NO ADJUSTMENTS.
4. WIND PRESSURES INDICATED ARE STRENGTH LEVEL VALUES.
5. TABLE LEGEND:
   - (+) = POSITIVE (INWARD) PRESSURE
   - (-) = NEGATIVE (OUTWARD) PRESSURE
   - SF = SQUARE FEET
6. FOR EFFECTIVE MEMBER AREAS NOT SPECIFICALLY LISTED, INTERPOLATE OR USE LARGEST VALUE OF WIND PRESSURE / SUCTION NOTED. DO NOT USE 1/3 STRESS INCREASE FOR MEMBER DESIGN WITH VALUES NOTED IN THIS TABLE.



FLAT ROOF
ROOF SLOPE 7° OR LESS

UNCOATED TENSION DEVELOPMENT & CLASS "B" LAP SPLICE SCHEDULE (f'c = 4,000 psi)

BAR SIZE	TENSION DEVELOPMENT LENGTH CLR COV = .75" BOT BARS	TOP BARS	CLR COV = 1" BOT BARS	TOP BARS	CLR COV = 1.5" BOT BARS	TOP BARS	CLASS "B" TENSION LAP LENGTH CLR COV = .75" BOT BARS	TOP BARS	CLR COV = 1" BOT BARS	TOP BARS	CLR COV = 1.5" BOT BARS	TOP BARS
#3	12	12	12	12	12	12	12	15	12	15	12	15
#4	15	19	12	15	12	15	19	24	15	20	15	20
#5	21	28	17	22	15	19	28	36	22	29	19	24
#6	29	37	24	31	17	22	37	48	31	40	22	29
#7	46	60	38	50	28	37	60	78	50	64	37	48
#8	57	74	48	62	36	47	74	96	62	80	47	60
#9	69	90	58	76	44	57	90	117	76	98	57	74
#10	83	108	70	92	54	70	108	140	92	119	70	91
#11	98	127	83	108	64	84	127	165	108	141	84	109

SCHEDULE NOTES:
1) BASED ON:
   1a. GRADE 60 REINFORCEMENT BARS.
   1b. NORMAL WEIGHT CONCRETE.
   1c. FOR BARS IN WALLS AND SLABS.
2) TOP BARS ARE HORIZONTAL BARS WITH MORE THAN 12" OF CONCRETE BELOW THE BARS.
3) FOR LIGHTWEIGHT CONCRETE, MULTIPLY TABLED VALUES BY 1.33.

THIS SCHEDULE IS PROVIDED FOR THE CONVENIENCE OF THE CONTRACTOR AND IS NOT INTENDED TO COVER ALL SITUATIONS. SHOP DRAWINGS SHALL CLEARLY INDICATE ALL REQUIRED LAP LENGTHS.

UNCOATED TENSION DEVELOPMENT & CLASS "B" LAP SPLICE SCHEDULE (f'c = 3,000 psi)

BAR SIZE	TENSION DEVELOPMENT LENGTH CASE 1 BOT BARS	TOP BARS	CASE 2 BOT BARS	TOP BARS	CLASS "B" TENSION LAP LENGTH CASE 1 BOT BARS	TOP BARS	CASE 2 BOT BARS	TOP BARS
#3	17	22	25	32	22	28	32	42
#4	22	29	33	43	29	37	43	56
#5	28	36	41	54	36	47	54	70
#6	33	43	50	64	43	56	64	84
#7	48	63	72	94	63	81	94	122
#8	55	72	82	107	72	93	107	139
#9	62	81	93	121	81	105	121	157
#10	70	91	105	136	91	118	136	177
#11	78	101	116	151	101	131	151	196

SCHEDULE NOTES:
1) BASED ON:
   1a. GRADE 60 REINFORCEMENT BARS.
   1b. NORMAL WEIGHT CONCRETE.
2) CASE DEFINITIONS (db = BAR DIAMETER)
   CASE 1: BEAMS, COLUMNS: W/ COVER ≥ 1.0db & C-C BAR SPACING ≥ 2.0db
   ALL OTHER UNO: W/ COVER ≥ 1.0db & C-C BAR SPACING ≥ 3.0db
   CASE 2: BEAMS, COLUMNS: W/ COVER < db OR C-C BAR SPACING < 2.0db
   ALL OTHER UNO: W/ COVER < db OR C-C BAR SPACING < 2.0db
3) TOP BARS ARE HORIZONTAL BARS WITH MORE THAN 12" OF CONCRETE BELOW THE BARS.
4) FOR LIGHWEIGHT CONCRETE, MULTIPLY TABLED VALUES BY 1.33.

THIS SCHEDULE IS PROVIDED FOR THE CONVENIENCE OF THE CONTRACTOR AND IS NOT INTENDED TO COVER ALL SITUATIONS. SHOP DRAWINGS SHALL CLEARLY INDICATE ALL REQUIRED LAP LENGTHS.

STANDARD ABBREVIATIONS:

AB	ANCHOR BOLT (ROD)	LLBB	LONG LEG BACK TO BACK
AHU	AIR HANDLING UNIT	LLH	LONG LEG HORIZONTAL
ALT	ALTERNATE	LLV	LONG LEG VERTICAL
APPROX	APPROXIMATELY	LP	LOW POINT
ARCH	ARCHITECTURAL	LSB	CLASS 'B' BAR LAP
BF	BOTTOM OF FOOTING	LSL	LAMINATED STRAND LUMBER
BS	BOTTOM OF STEEL	LTWT	LIGHTWEIGHT
BC	BOTTOM CHORD	LVL	LAMINATED VENEER LUMBER
BLDG	BUILDING	LW	LONG WAY
BRG	BEARING	MAX	MAXIMUM
BTWN	BETWEEN	MECH	MECHANICAL
CB	CATCH BASIN	MFR	MANUFACTURER
CIP	CAST-IN-PLACE	MIN	MINIMUM
CJ	CONTROL JOINT	MISC	MISCELLANEOUS
CL	CENTER LINE	MO	MASONRY OPENING
CLR	CLEAR (DISTANCE)	MS	MIDDLE STRIP
CMU	CONCRETE MASONRY UNIT	NA	NOT APPLICABLE
COL	COLUMN	NIC	NOT IN CONTRACT
CONC	CONCRETE	NOM	NOMINAL
CONT	CONTINUOUS	NTS	NOT TO SCALE
CS	COLUMN STRIP	OC	ON CENTER
DBA	DEFORMED BAR ANCHOR OR DECK BEARING ANGLE	OD	OUTSIDE DIAMETER
		OF	OUTSIDE FACE
DBE	DECK BEARING ELEVATION	OPNG	OPENING
DEMO	DEMOLITION / DEMOLISH	OPP	OPPOSITE
DIA	DIAMETER	OSL	OUTSTANDING LEG
DL	DEAD LOAD	PC	PRECAST / PRESTRESSED
DWG	DRAWING	PCI	POUNDS PER CUBIC INCH
EOD	EDGE OF DECK	PDF	POUNDS PER CUBIC FOOT
EOS	EDGE OF SLAB	PL	PLATE
EF	EACH FACE	PLBG	PLUMBING
EJ	EXPANSION JOINT	PLF	POUNDS PER LINEAR FOOT
EL	ELEVATION	PROJ	PROJECTION
ELEC	ELECTRICAL	PSF	POUNDS PER CUBIC FOOT
ENG	ENGINEER	PSI	POUNDS PER SQUARE INCH
EQ	EQUAL	PT	PRE (POST) -TENSIONED
ES	EDGE STRIP	RD	ROOF DRAIN
EW	EACH WAY	REF	REFERENCE
EWEF	EACH WAY EACH FACE	REINF	REINFORCE(D)
EXP	EXPANSION	REM	REMAINDER
EXT	EXTERIOR	RTU	ROOF TOP UNIT
EXTG or (e)	EXISTING	SC	SLIP CRITICAL
FD	FLOOR DRAIN	SCHED	SCHEDULE
FLG	FLANGE	SHT	SHEET
FLR	FLOOR	SIM	SIMILAR
FND	FOUNDATION	SL	SNOW LOAD
FTG	FOOTING	SLBB	SHORT LEGS BACK TO BACK
FRMG	FRAMING	SOG	SLAB-ON-GRADE
FUT	FUTURE	SPA	SPAC(ES)(ED)(ING)
FV	FIELD VERIFY	SPEC	SPECIFICATION(S)
GA	GAUGE	SQ	SQUARE
GALV	GALVANIZED	SS	STAINLESS STEEL
GC	GENERAL CONTRACTOR	STD	STANDARD
GLULAM	GLUE-LAMINATED BEAM(S)	SW	SHORT WAY
GT	GIRDER TRUSS	TF	TOP OF FOOTING
HK	HOOK	TL	TOP OF LEDGE
HORIZ	HORIZONTAL	TP	TOP OF PIER
HP	HIGH POINT	TS	TOP OF STEEL
HVAC	HEATING, VENTILATING, AND AIR CONDITIONING	TW	TOP OF WALL
		TC	TENSION CONTROL
HWS	HEADED WELDED STUD(S)	TC	TOP CHORD
ID	INSIDE DIAMETER	THK	THICK (NESS) (ENED)
IF	INSIDE FACE	TL	TOTAL LOAD
INT	INTERIOR	TYP	TYPICAL
JBE	JOIST BEARING ELEVATION	UNO	UNLESS NOTED OTHERWISE
K	KIP	VERT	VERTICAL
KO	KNOCKOUT PANEL	VIF	VERIFY IN FIELD
KSI	KIPS PER SQUARE INCH	VWA	VERIFY WITH ARCHITECT
L	ANGLE	WL	WIND LOAD
LB	POUNDS	WP	WORKING POINT
LL	LIVE LOAD	WWF	WELDED WIRE FABRIC

© 2026 Partners in Design Architects, Inc.

raSmith
CREATIVITY BEYOND ENGINEERING
project number 2220439
167.45 W. Bluemound Road
Brookfield, WI 53005-5938
(262) 781-1000
rasmith.com

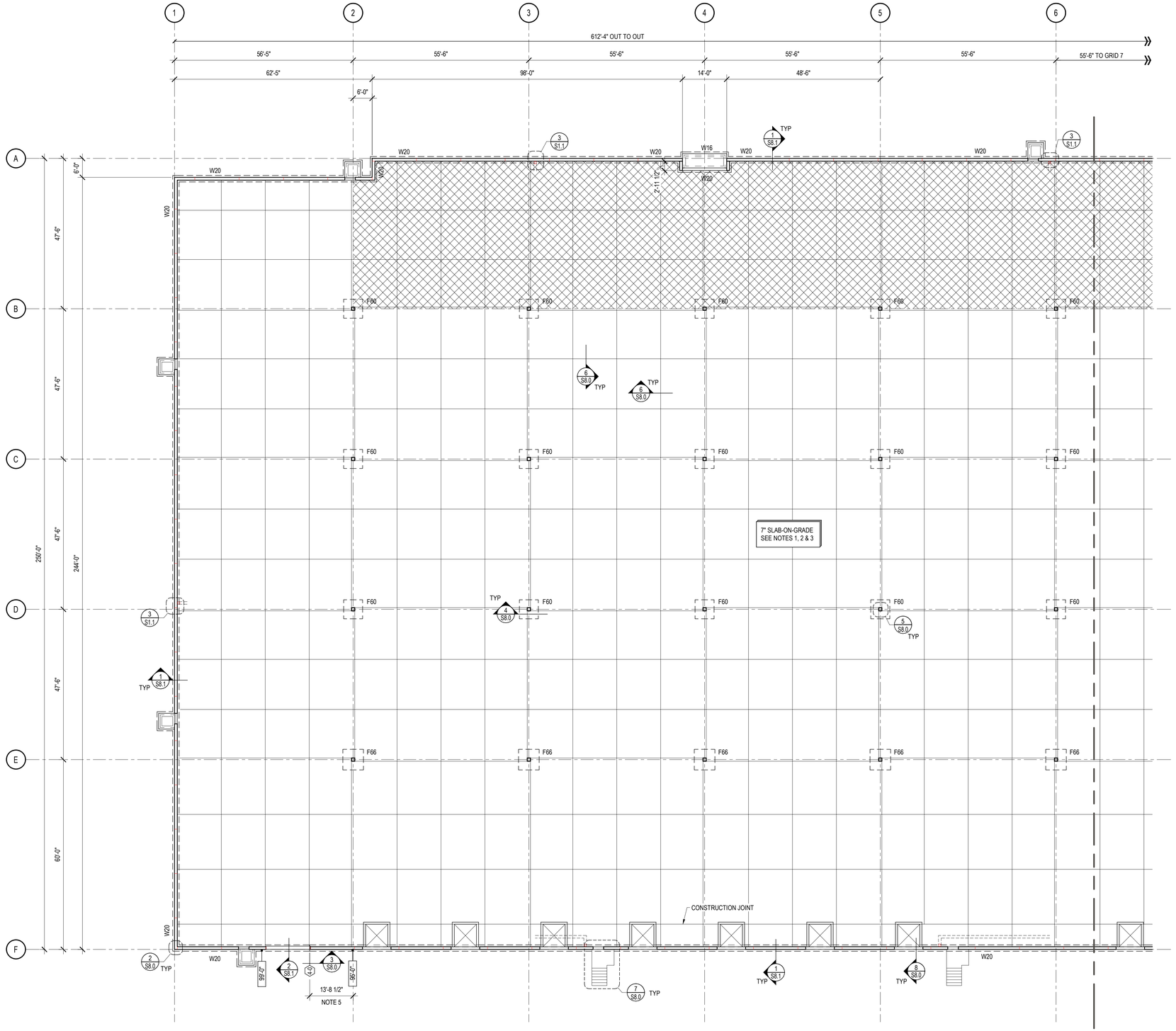
REVISIONS:
02/27/26 SITE & OPERATIONAL CONDITIONAL APPROVAL SET

ZILBER INDUSTRIAL 3
108th St, Lakeview Corporate Park, Pleasant Prairie, WI

STRUCTURAL NOTES

Partners in Design ARCHITECTS
2610 Lake Cook Road Suite 280 Riverwoods, IL 60015 Ph.: (847)940-0300

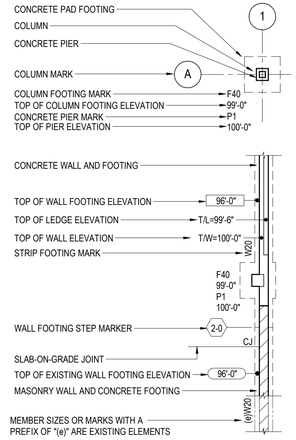
PROJECT NO.: 522.22.031
DRAWN BY: EMS CHECKED BY: SJR
DATE: 03.09.26
SHEET NO.: S0.2



1 S1.0 FOUNDATION PLAN - AREA A

SCALE: 1/16" = 1'-0"

FOUNDATION LEGEND



FOUNDATION PLAN NOTES

1. FINISH SLAB ELEVATION = 100'-0". LOCAL DATUM UNLESS NOTED OTHERWISE. TOP OF PERIMETER FOOTING ELEVATION = 96'-6" UNLESS NOTED OTHERWISE. TOP OF INTERIOR FOOTING ELEVATION = 98'-8" UNLESS NOTED OTHERWISE.
2. PROVIDE 7" THICK UNREINFORCED SLAB-ON-GROUND ON 6" THICK STONE BASE MEETING THE GRADATION REQUIREMENTS OF ASTM D2940 ON PORTLAND STABILIZED SUB-BASE. HARD STEEL TROWEL FINISH MEETING FF35/FL25 (OVERALL) AND FF24/FL17 (LOCAL).
3. TYPICAL WHERE SLAB-ON-GROUND ABUTS WALL OR COLUMN, PROVIDE 1/4" x (SOG THICKNESS) PRE-SCORED REMOVABLE TOP STRIP ISOLATION BOARD.
4. OVER-EXCAVATION PER DETAIL 2/S1.1 MAY BE REQUIRED TO REMOVE EXISTING UNDOCUMENTED FILL AND UNSUITABLE BEARING SOIL.
5. COORDINATE EXACT DIMENSION OF WALL STEP WITH PRECAST WALL PANEL LAYOUT. SEE 3/S8.0
6. TYPICAL DETAILS THAT APPLY TO PLAN INCLUDE:
   - 1/S8.0 SPREAD FOOTING OVER LATERAL
   - 2/S8.0 CORNER REINFORCEMENT DETAIL
   - 3/S8.1 STOOP DETAIL
7. IN AREAS SHOWN SOG, PROVIDE 10 MIL POLY VAPOR BARRIER DIERECTLY BELOW SLAB-ON-GROUND. SEE ARCHITECTURAL.

ISOLATED FOOTING SCHEDULE

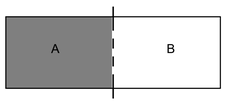
MARK	ISOLATED FOOTING DIMENSIONS			FOOTING REINFORCEMENT	REMARKS
	LENGTH	WIDTH	THICKNESS		
F60	6'-0"	6'-0"	17"	(8) #5; B, EW	
F66	6'-6"	6'-6"	18"	(6) #6; B, EW	

CONTINUOUS FOOTING SCHEDULE

MARK	CONTINUOUS FOOTING DIMENSIONS		FOOTING REINFORCEMENT	REMARKS
	WIDTH	THICKNESS		
W16	1'-6"	42"	(2) #6; T&B, CONT	AT STOOPS, SEE 3/S8.1
W20	2'-0"	42"	(3) #6 T&B, #3 TIES AT 48"OC	
W20a	2'-0"	42"	(3) #6 T&B, #3 TIES AT 48"OC	NOTE 3

NOTES:

1. B = BOTTOM, T = TOP, LW = LONG WAY, SW = SHORT WAY, EW = EACH WAY.
2. ALL REINFORCEMENT BARS TO BE BOTTOM BARS UNLESS NOTED OTHERWISE.
3. ALTERNATIVELY FOR W20a, PROVIDE 12" THICK FOOTING WITH (2) #5 BOTTOM BARS. IN EITHER CASE DROP BOTTOM OF FOOTING TO MATCH BOTTOM OF EXTERIOR WALL TRENCH FOOTING, HOLDING TOP OF FOOTING = 99'-0".
4. CONTINUOUS TRENCH FOOTING BARS CAN BE COUNTED FOR SPREAD FOOTING BARS AS LONG AS THEY ARE CONTINUOUS THROUGHOUT SPREAD FOOTING.



KEY PLAN
NOT TO SCALE

TRUE NORTH
PLAN NORTH

raSmith
CREATIVITY BEYOND ENGINEERING
project number: 2220438
167.45 W. Bluemound Road
Brookfield, WI 53005-5938
(262) 781-1000
rasmith.com

REVISIONS:
02/27/26 SITE & OPERATIONAL CONDITIONAL APPROVAL SET

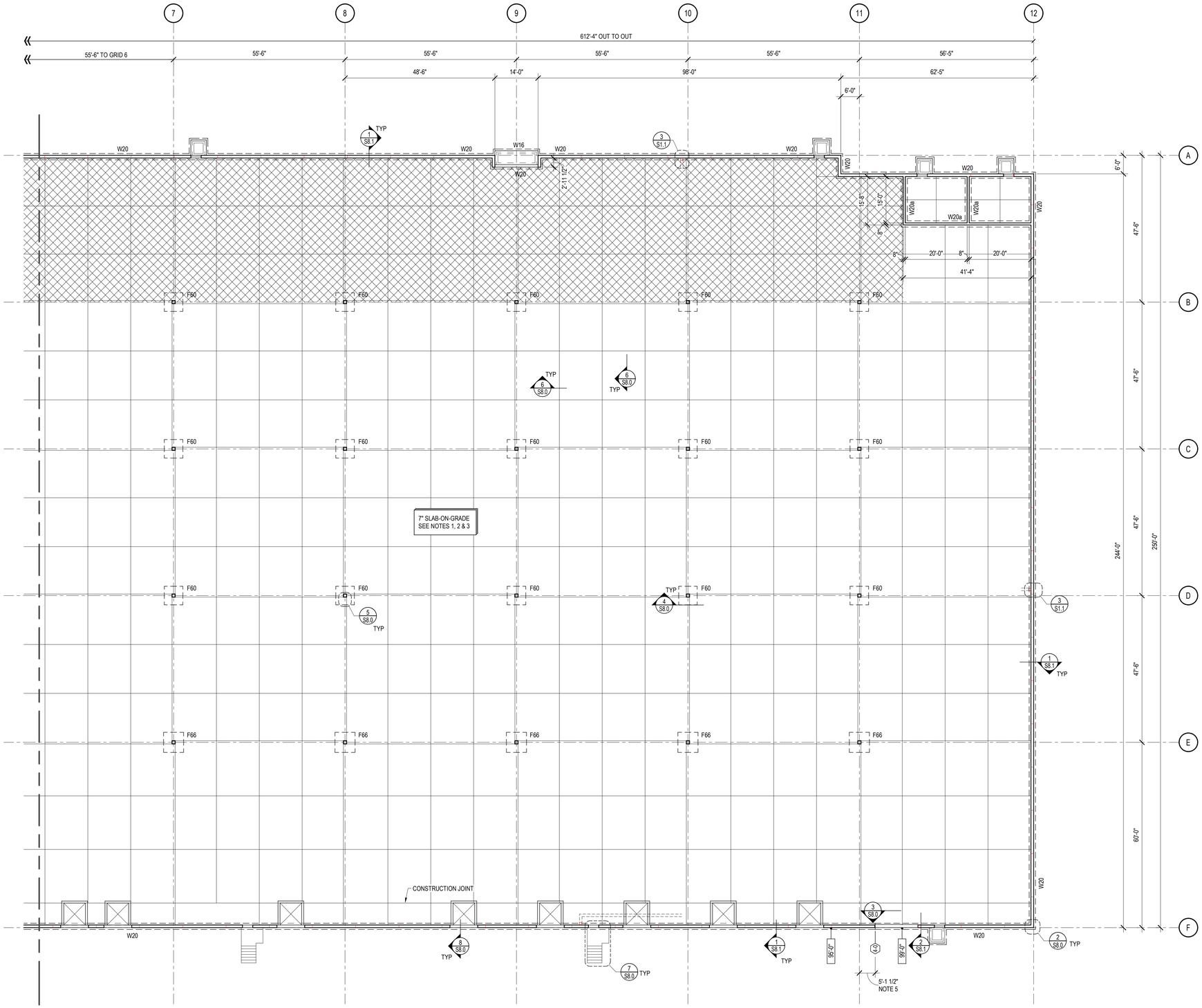
ZILBER INDUSTRIAL 3
108th St, Lakeview Corporate Park, Pleasant Prairie, WI

FOUNDATION PLAN

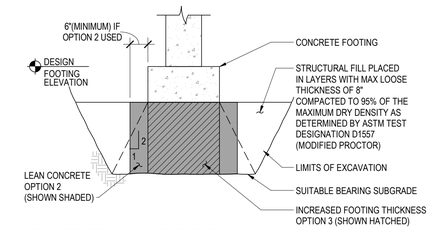
2610 Lake Cook Road Suite 280 Riverwoods, IL 60015 Ph.: (847)940-0300
Partners in Design ARCHITECTS

PROJECT NO.: 522.22.031
DRAWN BY: EMS CHECKED BY: SJR
DATE: 03.09.26
SHEET NO.: S1.0

© 2026 Partners in Design Architects, Inc.



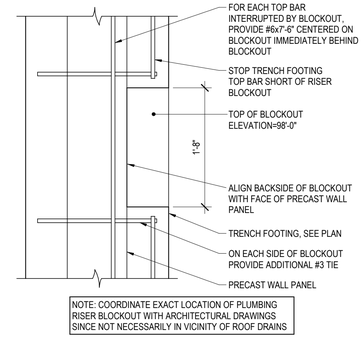
1 S1.1 FOUNDATION PLAN - AREA B
SCALE: 1/16" = 1'-0"



NOTE: CONTRACTOR MAY ELIMINATE STRUCTURAL FILL BY:
1. LOWERING DESIGNED FOOTING ELEVATION SO THAT FOOTING RESTS DIRECTLY ON SUITABLE BEARING SUBGRADE.
2. PROVIDING LEAN CONCRETE (f'c = 1,000 PSI) UNDER THE FOOTING TO SUITABLE BEARING SUBGRADE AS SHOWN SHADED ABOVE (NOT APPLICABLE IN GRANULAR SOILS).
3. INCREASING FOOTING THICKNESS TO REACH SUITABLE BEARING SUBGRADE AS SHOWN HATCHED ABOVE.

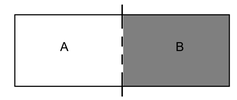
THIS DETAIL APPLIES ONLY AT THOSE LOCATIONS WHERE GEOTECHNICAL ENGINEER DEEM SOILS AT DESIGN FOOTING ELEVATIONS INADEQUATE FOR FOOTING SUPPORT. WHERE THIS WORK IS REQUIRED, CONTRACTOR WILL BE COMPENSATED ON A PRE-ESTABLISHED UNIT COST AGREED UPON BY ALL PRIOR TO THE WORK BEING COMPLETED.

2 S1.1 OVER-EXCAVATION DETAIL
SCALE: 1/2" = 1'-0"



NOTE: COORDINATE EXACT LOCATION OF PLUMBING RISER BLOCKOUT WITH ARCHITECTURAL DRAWINGS SINCE NOT NECESSARILY IN VICINITY OF ROOF DRAINS

3 S1.1 PLUMBING RISER BLOCKOUT
SCALE: 1" = 1'-0"



KEY PLAN
NOT TO SCALE

© 2026 Partners in Design Architects, Inc.

raSmith
CREATIVITY BEYOND ENGINEERING
project number 2220439
167A S. W. Bluemound Road
Brookfield, WI 53005-5938
(262) 781-1000
rasmith.com

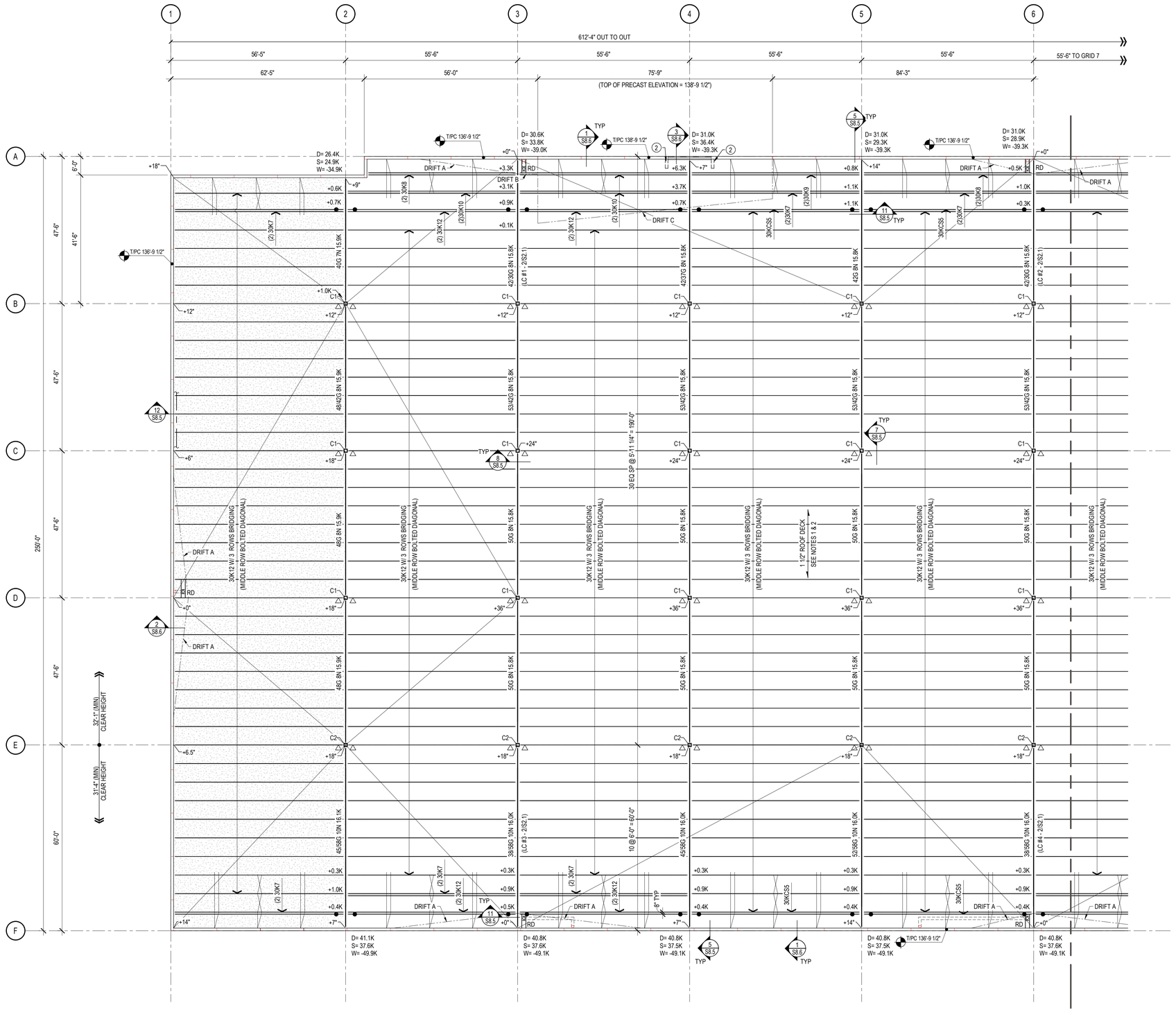
REVISIONS:
1 02.27.26 SITE + OPERATIONAL CONDITIONAL APPROVAL SET

ZILBER INDUSTRIAL 3
108th St, Lakeview Corporate Park, Pleasant Prairie, WI

FOUNDATION PLAN

2610 Lake Cook Road
Suite 280
Riverwoods, IL 60015
Ph.: (847)940-0300
Partners in Design ARCHITECTS

PROJECT NO.: 522.22.031
DRAWN BY: EMS
CHECKED BY: SJR
DATE: 03.09.26
SHEET NO.: S1.1

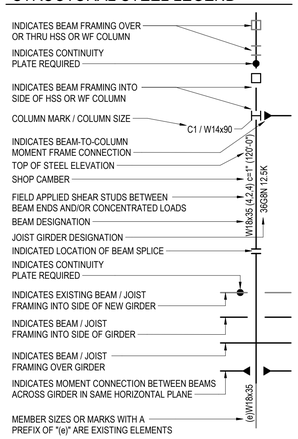


1 S2.0 ROOF FRAMING PLAN - AREA A

SCALE: 1/16" = 1'-0"

© 2026 Partners in Design Architects, Inc.

STRUCTURAL STEEL LEGEND



ROOF FRAMING PLAN NOTES

1. DECK BEARING ELEVATION @ '0' POINTS = 134'-10 1/2". ALL ELEVATIONS SHOWN ON PLAN ARE PLUS OR MINUS, AS INDICATED, FROM THIS DATUM ELEVATION. INCLUDING POINTS SHOWN OUTSIDE OF ROOF PLANE.
2. ROOF DECKING SHALL BE 1 1/2" x22 GA WIDE RIB GALVANIZED (G30) PRIME PAINTED WHITE METAL ROOF DECK FASTENED TO SUPPORTING STRUCTURE USING 36/4 PATTERN TYPICALLY OF FASTENING METHOD SHOWN ON 1/S8.5 AND #10 TEK SIDELAP FASTENERS AT 16" OC. PROVIDE DECK WITH THE FOLLOWING PROPERTIES:
   THICK = 0.0295 in $+M_d/\Omega$ = 422 #-FT/FT $+V_d/\Omega$ = 2654 LBS/FT
   Fy = 50 KSI $-M_d/\Omega$ = 477 #-FT/FT
   IN AREAS SHOWN [ ], USE 36/5 PATTERN W/ SIDELAPS AT 16" OC.
3. ▷ INDICATES LOCATION OF BAR JOIST BOTTOM CHORD EXTENSION. DO NOT WELD EXTENSION TO SUPPORT FRAMING. REFER TO DETAIL 8/S8.5.
4. ▶ INDICATES LOCATION OF BAR JOIST BOTTOM CHORD EXTENSION. WELD EXTENSION TO SUPPORT FRAMING AFTER ALL ROOF DEAD LOAD HAS BEEN APPLIED. SEE DETAIL 8/S8.5 AT JOIST GIRDERS.
5. JOIST SUPPLIER TO PROVIDE CONTINUOUS TOP AND BOTTOM CHORD HORIZONTAL ANGLE AND DIAGONAL X-BRIDGING WHERE SHOWN, NOTED AND AS REQUIRED. GC TO COORDINATE PLACEMENT OF JOIST BRIDGING WITH CLEARANCES REQUIRED BY FIRE PROTECTION LINES AND SPRINKLER HEADS.
6. ALL JOISTS TO BE CAMBERED IN ACCORDANCE WITH SJI STD SPECIFICATION UNLESS SPECIFICALLY NOTED OTHERWISE.
7. ALL BAR JOISTS TO BE DESIGNED FOR A NET WORKING UPLIFT LOAD OF 10 PSF IN ADDITION TO GRAVITY VERTICAL LOADS REQUIRED BY THE BAR JOIST DESIGNATION. A SINGLE LINE OF BOTTOM CHORD BRIDGING MUST BE PROVIDED NEAR THE FIRST BOTTOM CHORD PANEL POINTS.
8. DECK BEARING ANGLE BY PRECAST SUPPLIER, SEE 1/S8.6.
9. BAR JOIST BEARING ANGLE BY PRECAST SUPPLIER, SEE 2/S8.6.
10. ALL STEEL (BAR JOIST, BEAMS, AND COLUMNS) TO BE PRIME PAINTED GRAY.
11. WIND LOAD REACTIONS PROVIDED ON PLANS ARE STRENGTH VALUES (1.0W) UNLESS NOTED OTHERWISE.
12. PROVIDE ANGLE FRAME(S) SUPPORT AT ALL ROOF OPENINGS IN ACCORDANCE WITH DETAIL 3/S8.5 AT SMALLER OPENINGS, SEE 2/S8.5.
13. BELOW ROOFTOP UNIT CURB PROVIDE ANGLE STRUTS PER 4/S8.5 AND CHANNELS SHOWN PER 6/S8.5.
14. PROVIDE L3 1/2x2 1/2x1/4 (LLH) LAID ON TOP OF AND WELDED TO EACH JOIST ALONG EAST AND WEST WALLS PER 12/S8.5.
15. OUTER BAR JOISTS OF DOUBLE JOIST ACT AS CHORD FOR ROOF DIAPHRAGM, SEE 11/S8.5 FOR CONTINUITY PLATE REQUIRED AT EACH JOIST. (DO NOT ADJUST BAR JOIST SEAT HEIGHT)
16. HORIZONTAL SPANNING SPANDREL PANELS SHOWN [ ]. PRECAST SUPPLIER TO PROVIDE HAUNCH CONNECTION AND LATERAL RESTRAINT CONNECTIONS BETWEEN PANELS AS REQUIRED. BOTTOM OF SPANDREL ELEVATION PER PLAN. COORDINATE WITH ARCHITECTURAL DRAWINGS. SPANDREL IS 12" THICK OVERALL WITH BOTTOM AT ELEVATION = 116'-0".
17. SEE 1/S0.2 FOR DRIFTING SNOW LOAD DEFINITION.

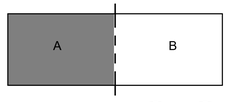
ROOF FRAMING KEY NOTES

1. PROVIDE 4'x4' ROOF HATCH OPENING, FRAME OPENING PER 3/S8.5.
2. TOP OF 10 1/2" PRECAST RETURN WALL PANELS BELOW AT ELEVATION 116'-4".

JG DESIGNATION NOTATION

32/44G8N16.8K (W = ± X, E = ± X)

- JOIST GIRDER DEPTH AT NORTH END
- JOIST GIRDER DEPTH AT SOUTH END
- AXIAL WIND & SEISMIC LOAD TO BE DESIGNED FOR (0.6W & 0.7E)



KEY PLAN
NOT TO SCALE

TRUE NORTH
PLAN NORTH

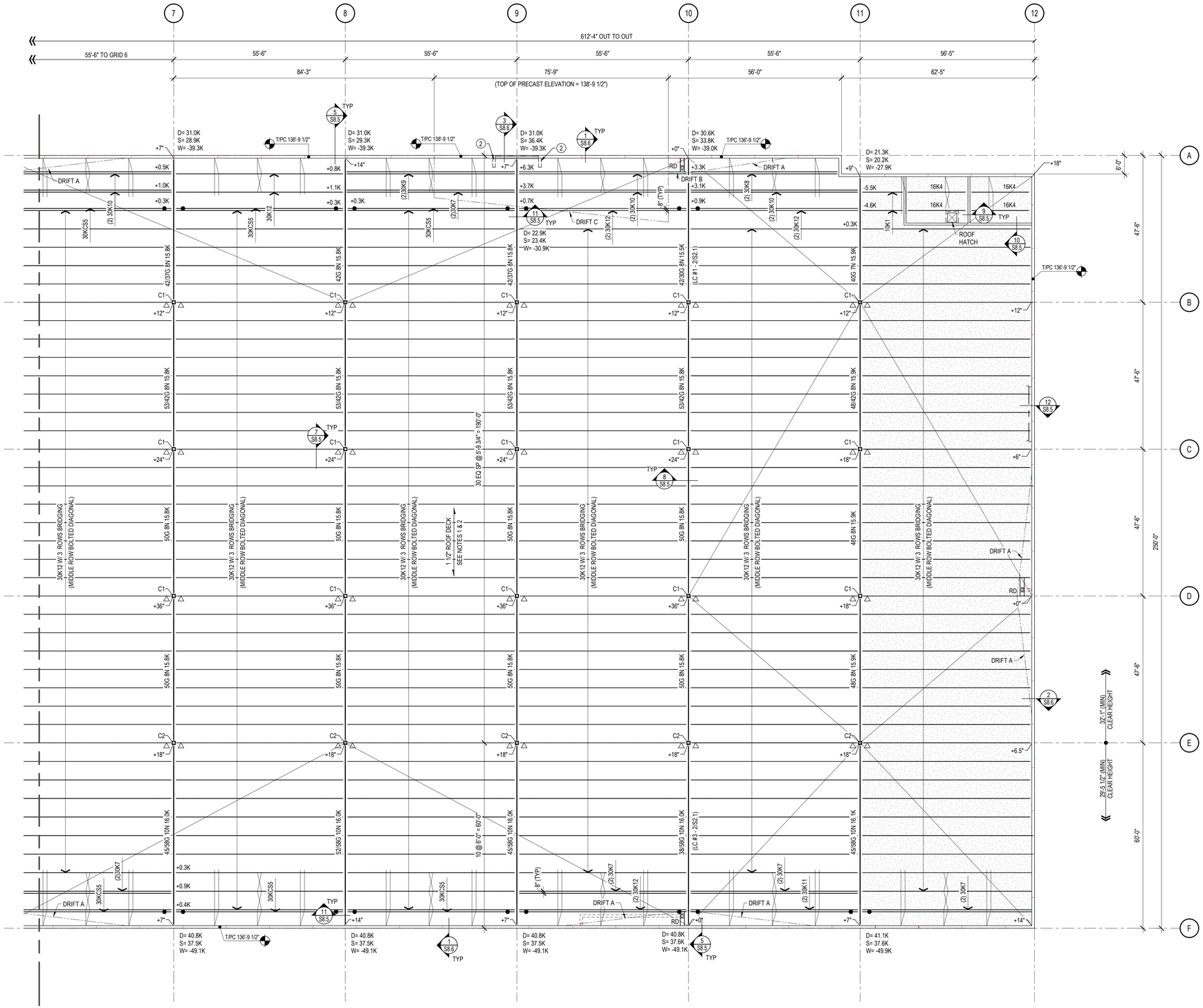
raSmith
CREATIVITY BEYOND ENGINEERING
project number: 222043
167.45 W. Bluemound Road
Brookfield, WI 53005-5938
(262) 781-1000
rasmith.com

REVISIONS:
02/27/26 SITE & OPERATIONAL CONDITIONAL APPROVAL SET

ZILBER INDUSTRIAL 3
108th St, Lakeview Corporate Park, Pleasant Prairie, WI
ROOF FRAMING PLAN

2610 Lake Cook Road
Suite 280
Riverwoods, IL 60015
Ph.: (847)940-0300
PartnersinDesign
ARCHITECTS

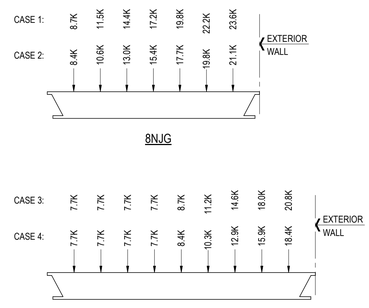
PROJECT NO.: 522.22.031
DRAWN BY: EMS
CHECKED BY: SJR
DATE: 03.09.26
SHEET NO.: S2.0



1 S2.1 ROOF FRAMING PLAN - AREA B

SCALE: 1/16" = 1'-0"

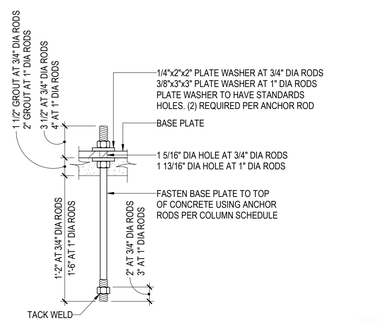
NOTE: BESIDES DESIGNING JOIST GIRDERS FOR LOAD BASED ON DESIGNATION, DESIGN JOIST GIRDERS FOR THESE SEPARATE LOAD CASES DUE TO PONDING WATER.



2 S2.1 JG SPECIAL LOAD CASE (PONDING WATER)

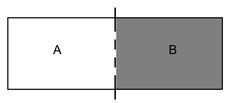
SCALE: 1/16" = 1'-0"

STEEL COLUMN SCHEDULE		
COLUMN MARK	C1	C2
CAP PLATE	7/S8.5	7/S8.5
COLUMN SIZE	HSS10x10x5/16	HSS12x12x3/8
BASE PLATE	3/4"x18"x1'-6"	3/4"x18"x1'-6"
ANCHOR BOLTS	(4) 3/4" DIA	(4) 3/4" DIA
DETAILS, NOTES	3/S2.1	3/S2.1



3 S2.1 STANDARD ANCHOR ROD DETAIL

SCALE: 1 1/2" = 1'-0"



KEY PLAN

NOT TO SCALE

TRUE NORTH

PLAN NORTH

raSmith
CREATIVITY BEYOND ENGINEERING
project number: 2220439
167A E. Bluemound Road
Brookfield, WI 53005-5938
(262) 781-1000
rasmith.com

REVISIONS:
02/27/26 SITE & OPERATIONAL CONDITIONAL APPROVAL SET

ZILBER INDUSTRIAL 3
108th St, Lakeview Corporate Park, Pleasant Prairie, WI

ROOF FRAMING PLAN

Partners in Design ARCHITECTS
2610 Lake Cook Road
Suite 280
Riverwoods, IL 60015
Ph.: (847)940-0300

PROJECT NO.: 522.22.031

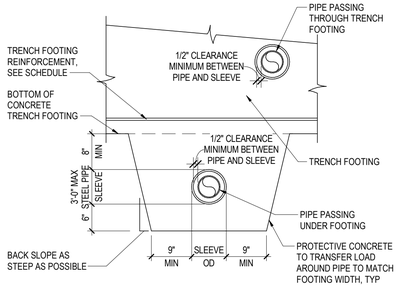
DRAWN BY: EMS

CHECKED BY: SJR

DATE: 03.09.26

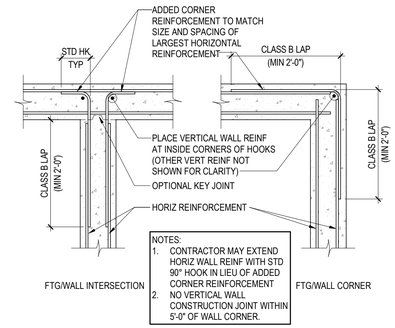
SHEET NO.: S2.1

© 2026 Partners in Design Architects, Inc.



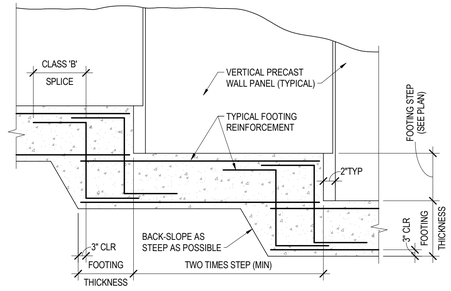
1 S8.0 PIPE PASSING UNDER/THROUGH WALL FOOTING

SCALE: 3/4" = 1'-0"



TYPICAL CONCRETE WALL AND FOOTING CORNER REINFORCEMENT

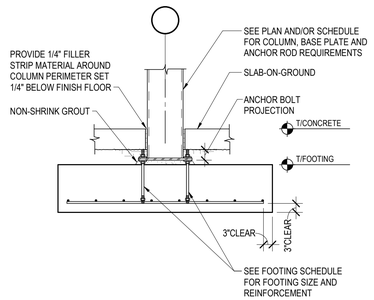
2 S8.0 SCALE: 3/4" = 1'-0"



NOTE:
COORDINATE FOOTING STEP LOCATIONS WITH PRECAST WALL PANEL LAYOUT

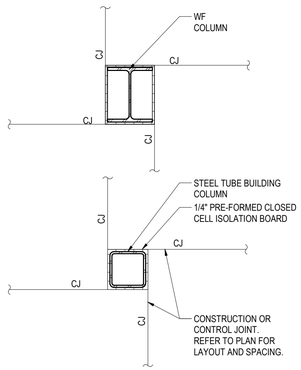
3 S8.0 FOOTING STEP DETAIL

SCALE: 1" = 1'-0"



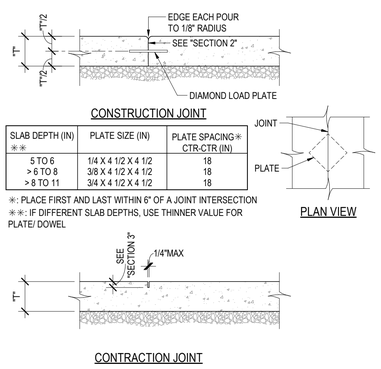
4 S8.0 TYPICAL INTERIOR COLUMN FOOTING

SCALE: 1/2" = 1'-0"



5 S8.0 JOINT LAYOUT AT COLUMN

SCALE: 3/4" = 1'-0"



SLAB DEPTH (IN) ✳✳	PLATE SIZE (IN)	PLATE SPACING✳ CTR-CTR (IN)
5 TO 6	1/4 X 4 1/2 X 4 1/2	18
> 6 TO 8	3/8 X 4 1/2 X 4 1/2	18
> 8 TO 11	3/4 X 4 1/2 X 4 1/2	18

✳: PLACE FIRST AND LAST WITHIN 6" OF A JOINT INTERSECTION
✳✳: IF DIFFERENT SLAB DEPTHS, USE THINNER VALUE FOR PLATE/ DOWEL

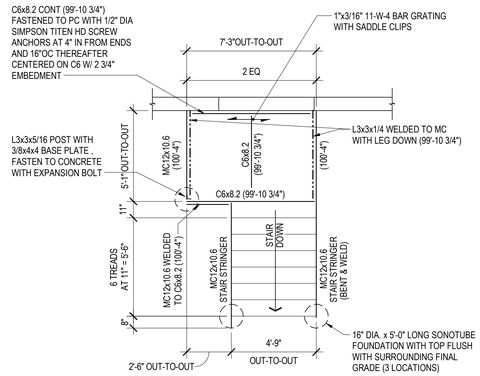
SECTION 1: SLAB-ON-GROUND NOTES
- SLAB-ON-GROUND TO BE PLACED USING ALTERNATING STRIPS IF SLAB NOT PLACED USING LASER SCREED.
- SLAB-ON-GROUND CONSTRUCTION SHOULD CONFORM WITH THE RECOMMENDATIONS AND REQUIREMENTS SET FORTH IN THE LATEST RELEASE OF ACI 302 GUIDE FOR CONCRETE FLOOR AND SLAB CONSTRUCTION.
- REFER TO THE GENERAL NOTES, THE SPECIFICATIONS, AND THE DRAWINGS FOR SUB-FLOOR DRAINAGE SYSTEM, SUBGRADE PREPARATION, AND/OR MUD SLAB AND VAPOR RETARDER REQUIREMENTS.
- THE SUBGRADE SHALL BE FREE OF STANDING WATER AT THE TIME OF CONCRETE PLACEMENT.
- REFER TO PLANS FOR SLAB THICKNESS ("T") AND REINFORCEMENT (WWF OR REINFORCEMENT BARS). REFER TO SPECIFICATIONS FOR FIBER REINFORCEMENT TO BE INCORPORATED IN CONCRETE MIX, IF ANY. WHERE PRESENT, REINFORCING BARS SHALL BE CHAIRED BY SOIL SUPPORTED SLAB BOLSTERS.
- PROVIDE (2) #5 x6'-0" AT ALL RE-ENTRANT CORNERS AND OTHER SIMILAR SLAB DISCONTINUITIES.
- UNLESS SHOWN OTHERWISE ON THE DRAWINGS, PROVIDE CONTRACTION AND/OR CONSTRUCTION JOINTS AT EVERY COLUMN LINE AND IN BETWEEN THE COLUMNS SUCH THAT THE JOINT SPACING DOES NOT EXCEED 30x("T") UNO. THE RESULTING PANELS SHOULD BE APPROXIMATELY SQUARE.

SECTION 2: CONSTRUCTION JOINT NOTES
- BREAK THE BOND BETWEEN NEW AND PREVIOUSLY PLACED SLABS BY SPRAYING OR BY PAINTING THE EXPOSED SIDE OF THE JOINT WITH A CURING COMPOUND, ASPHALTIC EMULSION, OR FORM OIL.

SECTION 3: CONTRACTION JOINT NOTES
- FOR SAW-CUT CONTRACTION JOINTS, MAKE THE SAW-CUT AS SOON AS THE SLAB IS ABLE TO SUPPORT THE WEIGHT OF WORKERS AND SAWING EQUIPMENT WITHOUT DAMAGE TO THE FINISHED SURFACE OF THE SLAB, BUT WITHIN 24 HOURS.
- DEPTH OF SAW-CUT SHOULD BE 1 1/2" IF PRODUCED USING THE EARLY ENTRY DRY-CUT PROCESS AND "T"/3 (1 1/2" MIN) IF PRODUCED USING THE CONVENTIONAL WET-CUT PROCESS.
- REFER TO SPECIFICATIONS REGARDING EPOXY RESIN OR ELASTOMERIC SEALANT FOR REQUIREMENTS AT CONTRACTION JOINTS.

6 S8.0 TYPICAL CONSTRUCTION AND CONTROL JOINTS IN SLAB-ON-GROUND

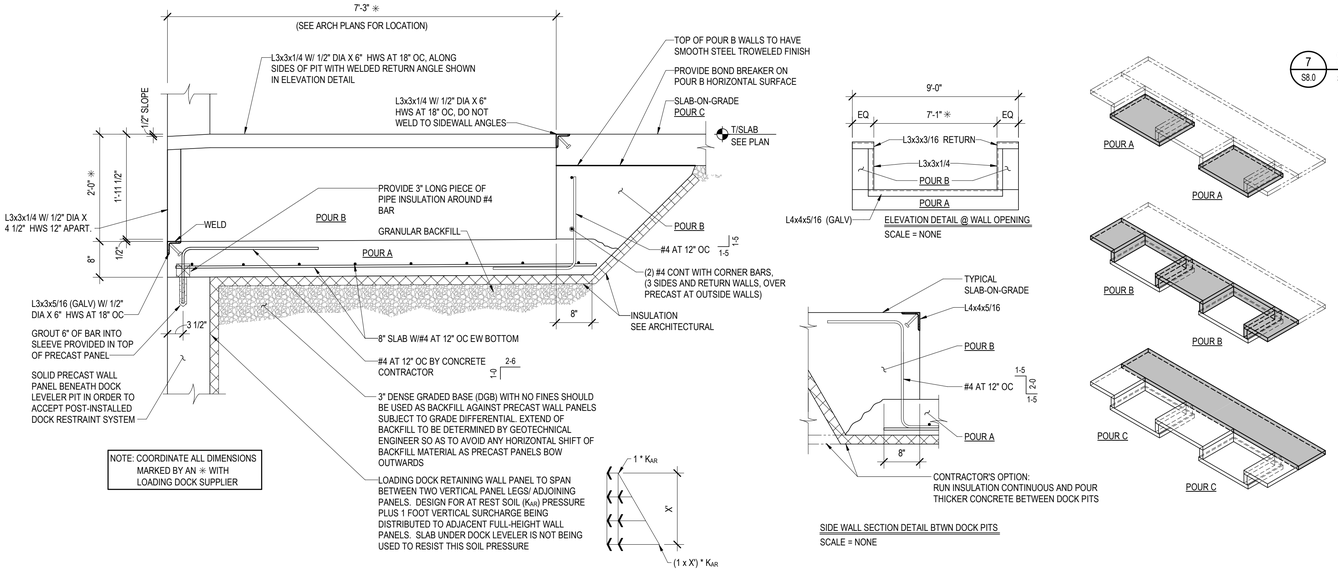
SCALE: 1" = 1'-0"



NOTES:
1) PROVIDE RAILING AT PLATFORM AND STAIRS PER ARCHITECTURAL DRAWINGS AND BUILDING CODE REQUIREMENTS.
2) FRAMING, GRATING, TREADS AND FASTENERS TO BE GALVANIZED.
3) IF C12x20.7 SECTIONS USED, HOLD INSIDE DIMENSIONS.

7 S8.0 STAIR FRAMING PLAN

SCALE: 1/4" = 1'-0"



SIDE WALL SECTION DETAIL BTWN DOCK PITS

SCALE = NONE

8 S8.0 LOADING DOCK LEVELER PIT

SCALE: 3/4" = 1'-0"

© 2026 Partners in Design Architects, Inc.

raSmith
CREATIVITY BEYOND ENGINEERING
project number 2220439
167.45 W. Bluemound Road
Brookfield, WI 53005-5938
(262) 781-1000
rasmith.com

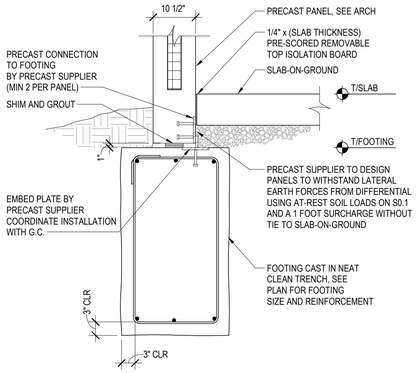
REVISIONS:
09.22.26 SITE & OPERATIONAL CONDITIONAL APPROVAL SET

ZILBER INDUSTRIAL 3
108th St, Lakeview Corporate Park, Pleasant Prairie, WI

FOUNDATION DETAILS

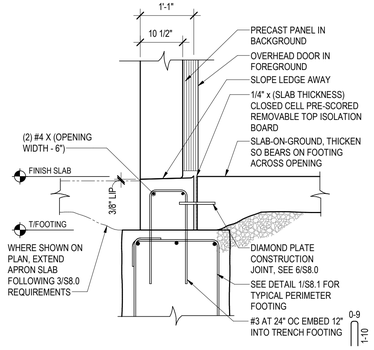
2610 Lake Cook Road
Suite 280
Riverwoods, IL 60015
Ph.: (847)940-0300
Partners in Design ARCHITECTS

PROJECT NO.: 522.22.031
DRAWN BY: EMS
CHECKED BY: SJR
DATE: 03.09.26
SHEET NO.: S8.0



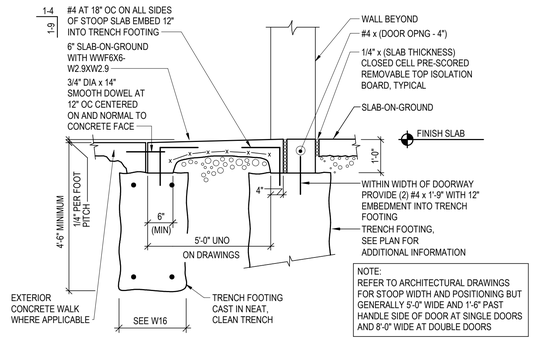
1 S8.1 TYPICAL PERIMETER FOOTING

SCALE: 3/4" = 1'-0"



2 S8.1 OVERHEAD DOOR THRESHOLD

SCALE: 3/4" = 1'-0"



3 S8.1 TYPICAL STOOP DETAIL

SCALE: 1" = 1'-0"

© 2026 Partners in Design Architects, Inc.

raSmith
CREATIVITY BEYOND ENGINEERING
project number 2220439
167.45 W. Bluemound Road
Brookfield, WI 53005-5938
(262) 781-1000
rasmith.com

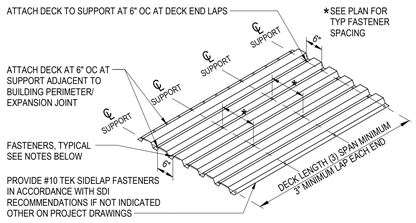
REVISIONS:
02/27/26 SITE & OPERATIONAL CONDITIONAL APPROVAL SET

ZILBER INDUSTRIAL 3
108th St, Lakeview Corporate Park, Pleasant Prairie, WI

FOUNDATION DETAILS

Partners in Design ARCHITECTS
2610 Lake Cook Road
Suite 280
Riverwoods, IL 60015
Ph.: (847)940-0300

PROJECT NO.: 522.22.031
DRAWN BY: EMS
CHECKED BY: SJR
DATE: 03.09.26
SHEET NO.: S8.1

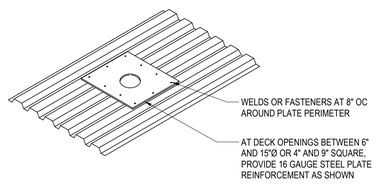


FAST DECK TO SUPPORTING MEMBERS USING FASTENING METHOD INDICATED ON DRAWINGS AND ACCORDING TO THE FOLLOWING REQUIREMENTS:

a. POWDER-ACTUATED FASTENERS: HILTI X-HSN 24 AND X-ENP 19L15, RAMSET 265D, OR APPROVED EQUAL (SUBMIT MANUFACTURER'S LITERATURE FOR APPROVAL ON EQUAL PRIOR TO USING POWDER-ACTUATED PRODUCT).

1 S8.5 METAL DECK FASTENING DETAIL

SCALE: 12" = 1'-0"



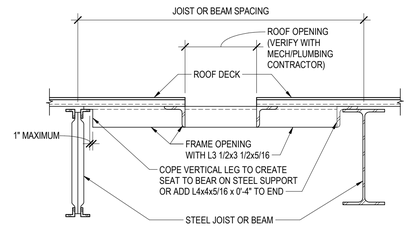
NOTE:

A. NO PLATE REINFORCEMENT REQUIRED AT DECK OPENINGS SMALLER THAN 6"Ø OR 4" SQUARE.

B. FOR DECK OPENINGS LARGER THAN 13"Ø OR 9" SQUARE, PROVIDE STEEL FRAME PER TYPICAL FRAMING AT ROOF OPENINGS DETAIL

2 S8.5 TYPICAL ROOF DECK OPENING

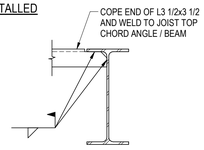
SCALE: 1" = 1'-0"



FOR FRAMES SET
BEFORE DECK INSTALLED

NOTE:
GC TO VERIFY SIZE, SHAPE AND LOCATION OF ALL ROOF OPENINGS WITH APPROPRIATE TRADES PROVIDING TYPICAL FRAME AT ALL OPENINGS:

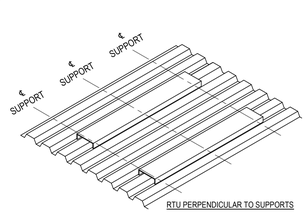
1. LARGER THAN 13" DIAMETER OR 9" SQUARE
   OR
2. SUPPORTING IN EXCESS OF 60 LBS EQUIPMENT LOADS

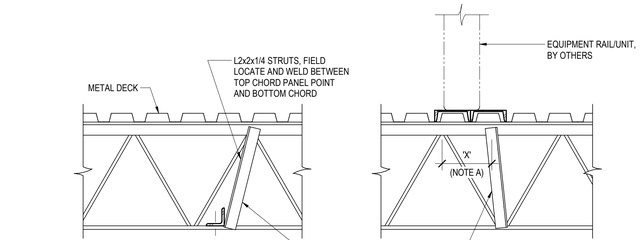


FOR FRAMES SET
AFTER DECK INSTALLED

3 S8.5 TYPICAL FRAMING AT ROOF OPENINGS

SCALE: 1" = 1'-0"





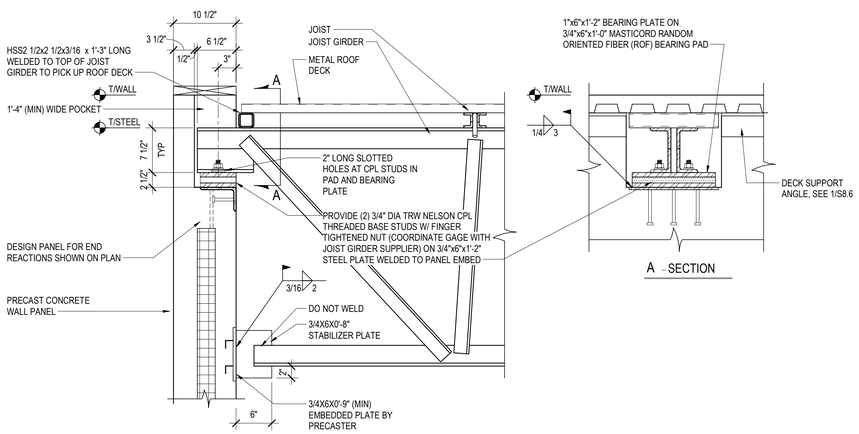
NOTE:
REINFORCEMENT REQUIRED AT ALL HANGING LOADS BETWEEN BOTTOM CHORD PANEL POINTS THAT EXCEED CRITERIA ON STRUCTUAL NOTES SHEET.

NOTE:
AS LONG AS CONCENTRATED LOAD ALONG JOIST TOP CHORD IS AT OR WITHIN 2" OF JOIST PANEL POINT OR CONCENTRATED LOAD ≤ 100# @ BOTTOM CHORD OR 300# @ TOP CHORD, L2x2 IS NOT REQUIRED.

NOTE:
PRIOR TO INSTALLATION, VERIFY LOADS WITH ENGINEER IF ACTUAL DIFFER FROM THOSE POSTED ON DRAWINGS.

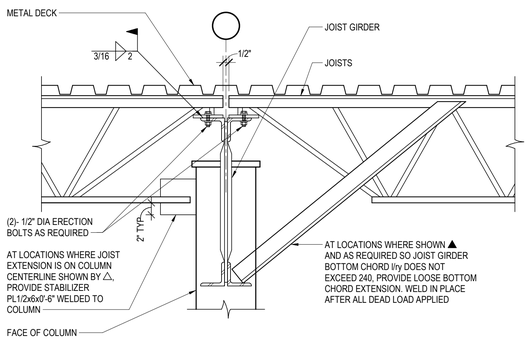
4 S8.5 CONCENTRATED LOAD AT JOIST

SCALE: 1" = 1'-0"



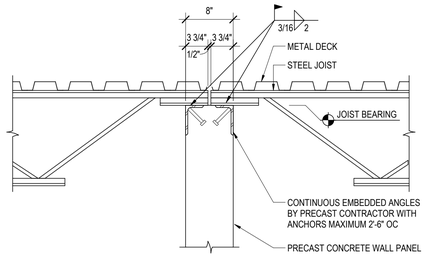
5 S8.5 JOIST GIRDER AT PRECAST PANEL

SCALE: 1" = 1'-0"



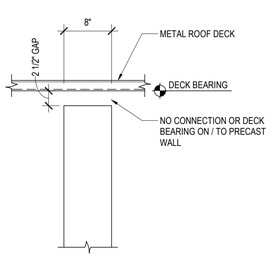
8 S8.5 TYPICAL JOIST-ON-JOIST GIRDER DETAIL

SCALE: 1" = 1'-0"



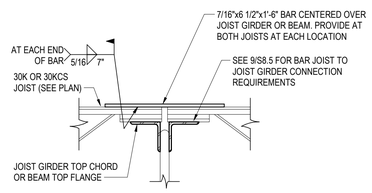
9 S8.5 SECTION

SCALE: 1" = 1'-0"



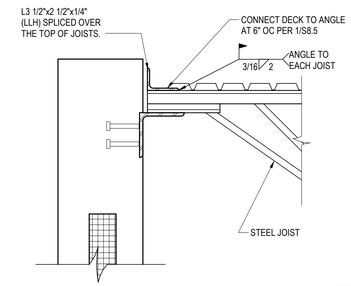
10 S8.5 SECTION

SCALE: 1" = 1'-0"



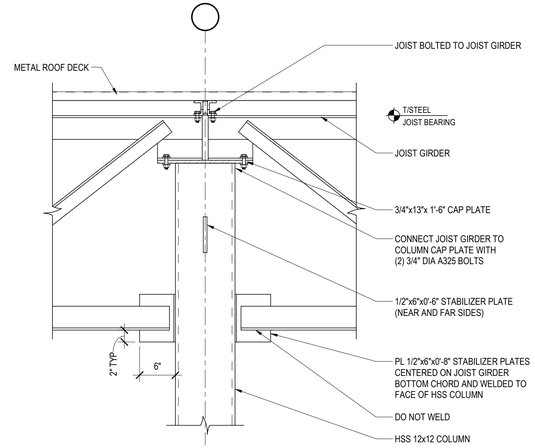
11 S8.5 BAR JOIST CONTINUITY PLATE

SCALE: 1" = 1'-0"



12 S8.5 ROOF DIAPHRAGM SHEAR COLLECTOR

SCALE: 1 1/2" = 1'-0"



7 S8.5 JOIST GIRDERS ON COLUMN

SCALE: 1" = 1'-0"

© 2026 Partners in Design Architects, Inc.

raSmith
CREATIVITY BEYOND ENGINEERING
project number: 2220438
167.45 W. Bluemound Road
Brookfield, WI 53005-5938
(262) 781-1000
rasmith.com

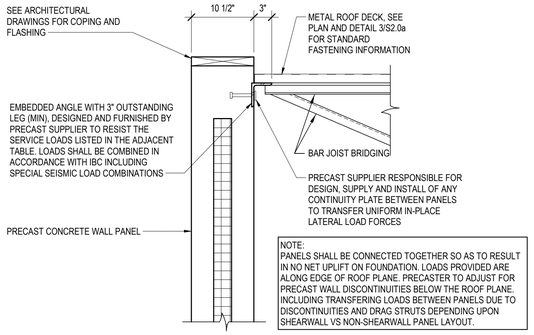
REVISIONS:
02.27.26 SITE & OPERATIONAL CONDITIONAL APPROVAL SET

ZILBER INDUSTRIAL 3
108th St, Lakeview Corporate Park, Pleasant Prairie, WI
FRAMING DETAILS

2610 Lake Cook Road
Suite 280
Riverwoods, IL 60015
Ph.: (847)940-0300

Partners in Design
ARCHITECTS

PROJECT NO.: 522.22.031
DRAWN BY: EMS
CHECKED BY: SJR
DATE: 03.09.26
SHEET NO.: S8.5

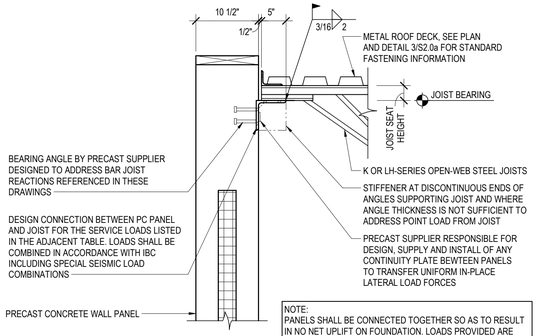


1 S8.6 JOIST BRIDGING AT PRECAST PANEL

SCALE: 1" = 1'-0"

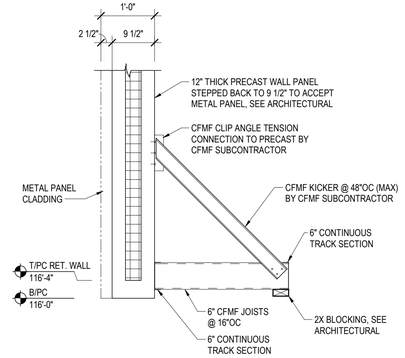
LOAD TABLE (PLF)				
DETAIL		1/S8.6		2/S8.6
TOP OF WALL ELEVATION		136'-9 1/2"	138'-9 1/2"	136'-9 1/2"
LOADS	VERTICAL (NOTE 1)			
	DEAD (DOWN)	78	78	685
	LIVE (DOWN)	75	201	700
	WIND (UP)	95	95	936
	HORIZONTAL PARALLEL TO WALL (NOTE 1)			
	WIND (+/-)	101	101	644
	SEISMIC (+/-)	132	132	430
	HORIZONTAL PERPENDICULAR TO WALL (NOTE 2)			
	WIND (+/-)	+546 / -511	+730 / -616	+546 / -511
	SEISMIC (+/-)	808	898	808

NOTES:
1. MULTIPLY BY BAR JOIST SPACING TO GET JOIST ROLL OVER FORCE AND POINT REACTION JOIST SUPPORT ANGLE IS TO BE DESIGNED FOR.
2. MULTIPLY BY BAR JOIST SPACING TO GET AXIAL LOAD TO BE TRANSFERRED THROUGH BAR JOIST SEAT ALONG DIRECTION OF JOIST SPAN.
3. WIND AND SEISMIC LOADS ARE STRENGTH LEVEL VALUES (1.0W, 1.0E)
4. LOADS PROVIDED ARE UNIFORM ALONG THE BACKSIDE OF PRECAST WALL PANEL AT ROOF PLANE. PRECASTER TO ADJUST FOR OPENINGS THAT OCCUR BELOW THE ROOF PLANE.



2 S8.6 JOIST AT PRECAST PANEL

SCALE: 1" = 1'-0"



3 S8.6 SECTION

SCALE: 3/4" = 1'-0"

© 2026 Partners in Design Architects, Inc.

raSmith
CREATIVITY BEYOND ENGINEERING
project number: 220439
1670 45 W. Bluemound Road
Brookfield, WI 53005-5938
(262) 781-1000
rasmith.com

REVISIONS:
02/27/26 SITE & OPERATIONAL CONDITIONAL APPROVAL SET

ZILBER INDUSTRIAL 3
108th St, Lakeview Corporate Park, Pleasant Prairie, WI
FRAMING DETAILS

2610 Lake Cook Road
Suite 280
Riverwoods, IL 60015
Ph: (847)940-0300
PartnersinDesign ARCHITECTS

PROJECT NO.: 522.22.031
DRAWN BY: EMS
CHECKED BY: SJR
DATE: 03.09.26
SHEET NO.: S8.6